Same Signal, Opposite Meaning: Direction-Aware Adaptive Compute for LLM Agents

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Abstract

¡¡¡¡¡¡¡ HEAD For LLM agents, *better-calibrated uncertainty signals can make adaptive test-time compute strictly worse than no gating at all*. Existing adaptive-gating methods share an implicit fixed-direction assumption—that higher uncertainty consistently indicates when extra computation helps—and we show this assumption fails in heterogeneous agent settings: the same signal value that predicts rollout benefit in one environment predicts rollout harm in another, and the sign can flip across model backbones with the task held fixed. When the assumed direction is wrong, improving calibration *worsens* performance by more precisely selecting harmful states—the gate becomes a more accurate harm-targeting mechanism. We trace this reversal to two coexisting state types with opposite signal–utility relationships—*information-poor* states, where rollouts cannot compensate for missing evidence, and *decision-difficult* states, where rollouts help explore viable alternatives—whose mixture is determined jointly by the environment and the backbone. The decomposition yields a sharper, information-theoretic claim: *the uncertainty signal is not a sufficient statistic for the value of computation*, so no policy depending on uncertainty alone—monotone or otherwise, calibrated or not—can deliver non-negative value across heterogeneous (environment, backbone) settings. Recovering sufficiency requires augmenting the signal with auxiliary features that distinguish the two state types. As an existence proof that this augmentation is the bottleneck—not gate sophistication—we instantiate the minimal procedure consistent with the analysis: DIAL (Direction-Informed Adaptive Learning), a sparse linear gate fit on signal-agnostic exploration data over a small pool of universal features. Across six environments and three backbones, DIAL already outperforms every fixed-direction baseline at zero per-step inference overhead, with its full range of gating behavior—from aggressive triggering to near-total suppression—emerging from direction learning alone. ======= Test-time compute improves decision quality for LLM agents, but deciding *when* to invoke it across diverse environments remains an open problem. Existing adaptive-gating methods assume that higher uncertainty consistently indicates when extra computation helps. We show this assumption fails: the same signal value that predicts rollout benefit in one environment predicts rollout harm in another, and the sign can flip across model backbones with the task held fixed. When the assumed direction is wrong, better-calibrated gates can *worsen* performance by more precisely selecting harmful states. We trace this reversal to a mixture of *information-poor* states, where rollouts cannot compensate for missing evidence, and *decision-difficult* states, where rollouts productively explore alternatives. Under this two-source model, uncertainty alone is insufficient to guarantee non-negative value across heterogeneous settings. Motivated by this, we propose DIAL (**D**irection-**I**nformed **A**daptive **L**earning), a sparse gate trained from signal-agnostic counterfactual exploration to learn the utility direction of state features. Across six environments and three backbones, DIAL consistently outperforms fixed-direction baselines. ¿¿¿¿¿¿¿ 1bf5b0e2c05afb339eb3e76f610bbf4ff034c5a3

1 Introduction

Large language models (LLMs) have rapidly advanced across a wide range of tasks, from question answering and code generation to web navigation and embodied decision making (Yao et al., 2023b; Shinn et al., 2023; Zhou et al., 2024). Yet, many tasks remain beyond the reach of a single forward pass, requiring the model to evaluate alternatives, verify intermediate results, or search over candidate actions (Yao et al., 2023a; Hao et al., 2023;

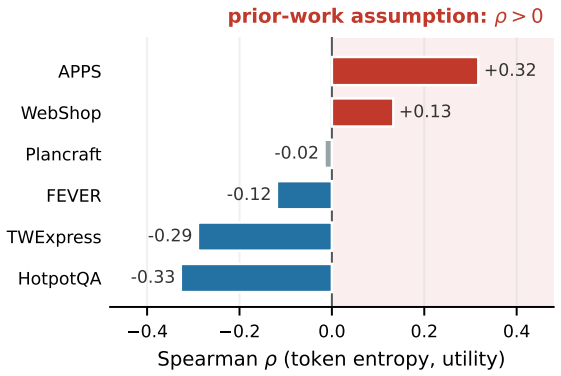

Figure 1: **The fixed-direction assumption fails on half of the environments—and the better the calibration, the worse the harm.** Prior entropy gating (Lee et al., 2025; Huang et al., 2026a; Jin et al., 2026) assumes $\rho(\text{entropy}, U)>0$ (shaded). On Qwen3-4B, 3 of 6 envs violate this (HotpotQA, TWExpress, FEVER), so high-entropy triggering fires on the wrong states. In these environments, more accurate calibration translates to more precise harm-targeting: reversing the sign of the gate's learned weights drops SR by 23–37 pp on strong-signal environments (Table 3).

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Figure 2: **The fixed-direction assumption fails on half of the environments.** Spearman correlation $\rho(\text{entropy}, U)$ across six environments on Qwen3-4B. Prior entropy gating (Lee et al., 2025; Huang et al., 2026a; Jin et al., 2026) assumes $\rho > 0$ (shaded region), but 3 of 6 environments violate this assumption.

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Snell et al., 2024). This has motivated *test-time compute*, where additional computation at inference time improves decision quality (Snell et al., 2024; Wang et al., 2026; Zhu et al., 2025a).

While effective, test-time compute is expensive, incurring substantial per-step overhead (e.g., requiring multiple rollouts or candidate evaluations at each decision point) (Yao et al., 2023a; Zhou et al., 2024). As agents are deployed across diverse settings, applying this overhead uniformly becomes wasteful when only a fraction of steps benefit. This has motivated *adaptive gating*: using uncertainty signals to decide *when* extra compute is worthwhile. Prior work explores diverse mechanisms, including entropy thresholding (Lee et al., 2025; Huang et al., 2026a; Jin et al., 2026), calibrated confidence (Muennighoff et al., 2025; Han et al., 2025), vote disagreement (Lee et al., 2026), and learned difficulty estimators based on hidden states or reinforcement learning (Liu et al., 2025; Zhu et al., 2025b; Zhang et al., 2025; Fang et al., 2025). ¡¡¡¡¡¡¡ HEAD

However, whether this assumption holds uniformly across heterogeneous agent environments has not been directly tested. Taking token entropy as the canonical signal adopted by most prior work, Figure 2 measures its correlation with optimizer utility across six agent environments: the common assumption is clearly violated on half of them, where high entropy instead indicates states on which extra compute is wasted. The signal–utility relationship therefore *reverses direction* across environments, with a sharp consequence: *the better an uncertainty-thresholding gate is calibrated, the worse it performs in mismatched settings*, because more accurate calibration translates to more precise harm-targeting (§5.3). This raises a central question: *if the signal–utility direction is not fixed, can adaptive compute still be principled?* ======= These methods differ in mechanism but share a common assumption: higher uncertainty consistently indicates when extra computation helps. We call this the *fixed-direction* assumption. Taking token entropy as the canonical signal, Figure 2 tests this assumption across six agent environments and finds it violated on half of them, where high entropy instead indicates states on which extra compute is wasted. The signal–utility relationship reverses direction across environments, with a sharp consequence: when the assumed direction is wrong, better calibration worsens performance by more precisely selecting harmful states. This raises a central question: *if the signal–utility direction is not fixed, can adaptive compute still be principled?* ¿¿¿¿¿¿¿
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¡¡¡¡¡¡¡ HEAD Extending the entropy view of Figure 2, we systematically measure the signal–utility relationship across six agent environments spanning QA (Yang et al., 2018), code generation (Hendrycks et al., 2021), web navigation (Yao et al., 2022), fact verification (Thorne et al., 2018), text games (Jansen and Côté, 2023), and crafting planning (Dagan et al., 2024), using three model backbones. The most striking finding is that *the reversal extends along the backbone dimension*: with the task held fixed, the sign of the entropy–utility correlation can flip across LLMs—FEVER, for instance, shifts from $\rho{=}{-}0.156$ on Phi-3.5 to $\rho{=}{+}0.428$ on Llama-3.1. The signal's semantics are therefore not a property of the environment alone but of the (environment, backbone) pair, ruling out any per-task tuning of di-

rection. Compounding this, the most informative signal itself also differs across settings, with structural features dominating in some environments while entropy carries little or no information in others. ======= Extending the entropy view of Figure 2, we systematically measure the signal–utility relationship across all six environments using three model backbones (Table 1). The most striking finding is that the reversal extends along the backbone dimension: with the task held fixed, the sign of the entropy–utility correlation can flip across LLMs. The signal's semantics are therefore not a property of the environment alone but of the (environment, backbone) pair, ruling out any per-task tuning of direction. Compounding this, the most informative signal itself also differs across settings, with structural features dominating in some environments while entropy carries little or no information in others. ¿¿¿¿¿¿¿ 1bf5b0e2c05afb339eb3e76f610bbf4ff034c5a3

¡¡¡¡¡¡¡ HEAD We trace this reversal to two coexisting state types with opposite signal–utility relationships. *Information-poor* states (Type I) arise when the agent lacks sufficient information; rollouts cannot synthesize information the environment itself does not provide, and high uncertainty predicts *negative* value of computation. In contrast, *decision-difficult* states (Type D) arise when multiple viable options exist; additional computation helps explore alternatives, and high uncertainty predicts *positive* value. Crucially, whether a given state is Type I or Type D depends on what the agent already encodes, so the mixture is determined jointly by the environment and the backbone—directly explaining why direction reverses along both axes. The same uncertainty signal carries opposite meanings depending on which type dominates, an instance of Simpson's paradox (Simpson, 1951; Wang et al., 2023a). This decomposition makes a falsifiable prediction—reversal is determined by the (environment, backbone) pair's mixture of Type I and Type D states—which we verify on held-out data (§5.4). It also yields a sharper, information-theoretic implication: *the uncertainty signal σ is not a sufficient statistic for the value of computation*, so no policy of the form $g(\sigma(s))$—monotone or otherwise, calibrated or not—can deliver non-negative value across heterogeneous (environment, backbone) settings (Proposition 1). Fixed-direction gating is the strict-monotone special case of a structurally insufficient family. ======= We trace this reversal to two coexisting state types with opposite signal–utility relationships. *Information-poor* states (Type I) arise when the agent lacks sufficient information; rollouts cannot synthesize what the environment does not provide, so high uncertainty predicts negative value of computation. For example, a fact-verification agent that hesitates because retrieved evidence is ambiguous gains nothing from additional rollouts. *Decision-difficult* states (Type D) arise when multiple viable options exist; rollouts help explore alternatives, so high uncertainty predicts positive value. For example, a coding agent that hesitates because several valid implementations exist benefits from exploring them.

Whether a given state is Type I or Type D depends on what the agent already encodes, so the mixture is determined jointly by the environment and the backbone. The same uncertainty signal carries opposite meanings depending on which type dominates, an instance of Simpson's paradox (Simpson, 1951; Wang et al., 2023a). This decomposition yields a formal implication: the uncertainty signal is not a sufficient statistic for the value of computation, so under the two-source model, no policy depending on uncertainty alone can deliver non-negative value across heterogeneous settings (Proposition 1). It also makes a falsifiable prediction, that reversal is determined by the mixture of Type I and Type D states, which we verify on held-out data (§5.4). Recovering sufficiency requires augmenting the uncertainty signal with features that correlate with the latent state type, such as step count as a proxy for information accumulation. ¿¿¿¿¿¿¿ 1bf5b0e2c05afb339eb3e76f610bbf4ff034c5a3

¡¡¡¡¡¡¡ HEAD As an existence proof that direction—not gate capacity—is the bottleneck, we instantiate the minimal direction-learning procedure consistent with this analysis: **DIAL** (**D**irection-**I**nformed **A**daptive **L**earning), a sparse linear gate fit on signal-agnostic exploration data. Across six environments and three LLM backbones, DIAL already outperforms every fixed-direction baseline at zero per-step inference overhead, and its full range of gating behavior—from aggressive triggering to near-total suppression—emerges from direction learning alone. The simplicity of the gate is the point: once the direction is learned, the simplest classifier suffices, indicating that the bottleneck for cross-setting adaptive compute lies in the assumption tested here rather

than in the sophistication of the gate. ======= As an existence proof that direction is a first-order bottleneck that simple gates can address once utility feedback is collected, we propose DIAL (**D**irection-**I**nformed **A**daptive **L**earning). DIAL collects unbiased state–utility pairs via randomized exploration, then fits a sparse linear gate over a small set of state features, optionally augmented by LLM-proposed task-specific signals, to learn the signal–utility direction for each (environment, backbone) setting. Across six environments and three LLM backbones, DIAL outperforms every fixed-direction baseline, with its gating behavior ranging from aggressive triggering to near-total suppression depending on the learned direction. Once the direction is correctly identified, even the simplest classifier suffices. ¿¿¿¿¿¿¿ 1bf5b0e2c05afb339eb3e76f610bbf4ff034c5a3

2 Related Work

¡¡¡¡¡¡¡ HEAD Prior work on adaptive test-time compute spans diverse mechanisms but rests on a shared, almost always implicit, assumption: that uncertainty signals have a fixed, setting-independent relationship with compute value. We organize the literature around how each line of work commits to this *fixed-direction* assumption, since it is precisely what this paper tests. ======= Prior work on adaptive test-time compute spans diverse mechanisms but shares a common assumption: that uncertainty signals have a fixed, setting-independent relationship with compute value. We organize the literature around how each line of work commits to this assumption. ¿¿¿¿¿¿¿ 1bf5b0e2c05afb339eb3e76f610bbf4ff034c5a3

Gating on uncertainty signals. Signal-based methods allocate extra computation when token confidence, entropy, or disagreement exceeds a threshold, encoding the fixed-direction assumption directly in their trigger rule. In reasoning settings, this takes the form of extended chain-of-thought generation (Lee et al., 2025; Huang et al., 2026a; Jin et al., 2026; Sharma and Chopra, 2025; Muennighoff et al., 2025; Han et al., 2025). In agent settings, vote-based methods (Lee et al., 2026) invoke an arbiter when candidate actions disagree, and verbalized uncertainty methods (Zhang et al., 2026) trigger reflection when self-reported confidence is low.

Gating on learned difficulty. Hidden-state probes (Liu et al., 2025; Zhu et al., 2025b) estimate task difficulty from internal representations, and RL-based methods (Zhang et al., 2025; Fang et al., 2025; Aggarwal and Welleck, 2025; Paglieri et al., 2025) learn think/no-think policies end-to-end. ¡¡¡¡¡¡¡ HEAD Although these methods commit to neither a specific signal nor an explicit threshold, their gating policies share the assumption that higher difficulty warrants more compute—identical in *direction* to signal-based approaches. The combined body of work therefore rests on the same implicit fixed-direction assumption, and is evaluated almost exclusively on homogeneous reasoning benchmarks (GSM8K (Cobbe et al., 2021), GPQA (Rein et al., 2023)) where the assumption holds by construction. We construct this assumption explicitly and test it in the heterogeneous agent setting, where it fails. ======= Although these methods commit to neither a specific signal nor an explicit threshold, their evaluations primarily focus on homogeneous reasoning benchmarks such as GSM8K (Cobbe et al., 2021) and GPQA (Rein et al., 2023), where higher difficulty is typically aligned with greater benefit from additional reasoning. As a result, they leave untested whether the difficulty–compute relationship itself can reverse. In contrast, we make this fixed-direction assumption explicit and test it in heterogeneous agent environments, where it fails.

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Metareasoning and uncertainty semantics. Rational metareasoning (Russell and Wefald, 1991; Sabbata et al., 2024) formalizes the value of computation (VOC), classically assuming VOC ≥ 0 since an agent can disregard unfavorable results. We show this assumption fails when the same model both proposes and evaluates actions (§3.2). Tao et al. (2025) and Heo et al. (2025) find that calibration quality varies across task types, and Abbasi Yadkori et al. (2024) decouple epistemic and aleatoric uncertainty, providing converging evidence that uncertainty semantics are domain-dependent. However, none observes that signal direction can reverse across environments. Our two-source model formalizes this domain dependence and makes it actionable. Extended related work discussion is in Appendix A.

Table 1: Signal–utility correlations across 6 environments and 3 backbones. ρ: Spearman correlation of token entropy with optimizer utility U. *: significant at $p<0.05$. Direction varies across *both* environments and model backbones.

Environment	Qwen3	Phi-3.5	Llama-3.1	Cons.
HotpotQA	-0.327^*	$+0.184^*$	-0.346^*	✗
WebShop	$+0.133^*$	$+0.335^*$	$+0.287^*$	✓
FEVER	-0.119^*	-0.156^*	$+0.428^*$	✗
APPS	$+0.317^*$	-0.024	-0.249^*	✗
TWExpress	-0.290^*	$+0.000$	$+0.000$	—
Plancraft	-0.016	$+0.000$	-0.176^*	—

3 Why Fixed-Direction Gating Fails

We first formalize the adaptive gating problem, then measure signal–utility correlations across diverse environments to test the fixed-direction assumption.

We study LLM-based agents in multi-step interactive environments. At each step t, the agent observes state s_t, selects action a_t, and transitions to s_{t+1}. An environment-specific *test-time optimizer* T (e.g., rollout evaluation, K-variant sampling) can be invoked at any step to potentially improve action quality. Let $\sigma(s)$ denote the uncertainty signal at state s (e.g., token entropy of the next-token distribution). We define the *optimizer utility* as:

$$U(T, s) = \mathbb{E}[R(\tau) \mid a{=}T(s)] - \mathbb{E}[R(\tau) \mid a{=}\pi(s)] \tag{1}$$

where $R(\tau)$ is the episode return. $U > 0$ indicates the optimizer improves the outcome; $U \leq 0$ indicates it is wasteful or harmful. The *adaptive gating problem* is to learn a gate $g : \mathcal{S} \to \{0, 1\}$ that triggers T only when $U > 0$.

3.1 Signal–Utility Direction Reverses Across Environments and Backbones

The signal–utility direction is not fixed. Table 1 reports Spearman correlations between token entropy and optimizer utility across 6 environments ¡¡¡¡¡¡¡ HEAD and 3 model backbones (Team, 2025, 2024; Microsoft, 2024). Two patterns stand out, and the more striking one is the harder to explain. First, *the same signal in the same environment can reverse across backbones*: in 3 out of 4 environments with multiple significant correlations, the backbones disagree on the sign of ρ. The most extreme case is FEVER, which flips from $\rho{=}-0.156$ on Phi-3.5 to $\rho{=}+0.428$ on Llama-3.1—a sign change driven ======= and 3 model backbones (Team, 2025, 2024; Microsoft, 2024). First, *the same signal in the same environment can reverse across backbones*: among environments with at least two significant nonzero correlations, sign disagreement across backbones is common. The most extreme case is FEVER, which flips from $\rho{=}-0.156$ on Phi-3.5 to $\rho{=}+0.428$ on Llama-3.1, a sign change driven ¿¿¿¿¿¿¿ 1bf5b0e2c05afb339eb3e76f610bbf4ff034c5a3 entirely by the choice of LLM, with the task held fixed. The signal's semantics are therefore not a property of the environment alone, but of how the agent's prior knowledge interacts with it. Second, *even within a single backbone, the direction also varies across environments*: on Qwen3, entropy is negatively correlated with utility in information-seeking environments (FEVER, HotpotQA, TWExpress) but positively in code generation (APPS) and web navigation (WebShop), with near-zero correlation in weak-signal environments (Plancraft). Crucially, this direction mismatch is not benign: as Section 5.2 will show, gates that assume the wrong direction systematically fall *below* the no-trigger baseline, making them strictly worse than doing nothing in mismatched (environment, backbone) pairs. ¡¡¡¡¡¡¡ HEAD

Implication: σ is not a sufficient statistic for VOC. The reversal in Table 1 is not just "fixed-direction gating fails"—it is the empirical signature of a stronger structural fact. Across (environment, backbone) settings, two states with identical σ values can have opposite-sign value of computation, depending on whether the agent is in an information-poor or decision-difficult state (§3.2). Hence σ is not a sufficient statistic for VOC, and *no* policy of the form $g(\sigma(s))$—monotone or otherwise, calibrated or not—can deliver non-negative value of computation across heterogeneous settings (Proposition 1, proof in Appendix B.1). Fixed-direction gating is the strict-monotone special case; the underlying obstruction is a property of the signal itself, not of how the gate post-processes it. Recovering sufficiency therefore requires augmenting σ with auxiliary features that distinguish the two state types—a requirement that directly motivates the multi-feature design of §4.

A diagnostic for practitioners. The reversal test of Table 1 is itself the leave-behind tool of this paper: collecting a few hundred signal-agnostic (s, U) pairs and reporting the Spearman correlation $\rho(\sigma, U)$ together with its boot-

strap confidence interval is sufficient to decide whether a fixed direction is safe for a given (environment, backbone) deployment. A confidence interval that crosses zero, or two settings that disagree on sign, indicates that any uncertainty-thresholding gate will eventually fall below never-gating; a stable single-sign correlation indicates that fixed-direction gating is safe and direction learning is unnecessary. We apply this diagnostic throughout, and recommend it as a standalone check independent of the specific learning procedure adopted. ======= ¿¿¿¿¿¿¿ 1bf5b0e2c05afb339eb3e76f610bbf4ff034c5a3

Signal identity also varies across environments. Beyond direction, the most informative signal itself differs across environments: `step_count` dominates in most settings, but WebShop relies on `num_available_actions` instead (Appendix F.1). No single signal is reliably informative; multi-signal gates substantially outperform single-feature gates (Appendix F.3). These observations suggest that uncertainty is entangled with latent state structure that varies across (environment, backbone) pairs. We formalize this intuition next.

3.2 Why Does Direction Reverse? A Two-Source Model

The findings above establish that fixed-direction gating fails; we ¡¡¡¡¡¡¡ HEAD now ask *why*. We trace direction reversal to two coexisting state types with opposite signal–utility relationships.

Intuition. Consider two agents that both hesitate (high entropy) at a decision point. A coding agent hesitates because multiple valid implementations exist; rollouts explore these alternatives productively, so high entropy positively predicts rollout utility (Type D). A fact-verification agent hesitates because the retrieved evidence is ambiguous; rollouts from the same model draw observations from the *same* information-bounded environment, so they cannot synthesize evidence the environment did not provide, and high entropy instead predicts rollout *futility* (Type I). The same numerical signal carries opposite meaning because the *source* of uncertainty differs. ======= now ask *why*. We model direction reversal as arising from two coexisting state types with opposite signal–utility relationships. As discussed in §1, high entropy carries opposite meaning depending on whether the agent lacks information (Type I, where rollouts cannot help) or faces multiple viable options (Type D, where rollouts productively explore alternatives). ¿¿¿¿¿¿¿ 1bf5b0e2c05afb339eb3e76f610bbf4ff034c5a3

Model. We model direction reversal as arising from two state types at each decision step:

- **Type I (information poverty):** Utility is *negatively* related to entropy: $U_I(s) \sim -\alpha H(s) + \varepsilon_I$, $\alpha > 0$.
- **Type D (decision difficulty):** Utility is *positively* related to entropy: $U_D(s) \sim +\beta H(s) + \varepsilon_D$, $\beta > 0$.

Let $p_I(\mathcal{E})$ denote the fraction of Type I states in environment $\mathcal{E}$. The marginal correlation becomes

$$\rho(\mathcal{E}) \;\approx\; \beta - (\alpha + \beta)\, p_I(\mathcal{E}), \qquad (2)$$

which reverses sign at the critical threshold $p_I^* = \beta/(\alpha + \beta)$. Environments with $p_I > p_I^*$ exhibit negative ρ (Type I dominated); those with $p_I < p_I^*$ exhibit positive ρ (Type D dominated); near p_I^*, the two effects cancel and entropy carries ¡¡¡¡¡¡¡ HEAD negligible signal. This prediction is consistent with the observed Qwen3 pattern in Table 1: search-based QA (FEVER, HotpotQA, TWExpress) shows negative ρ, code generation (APPS) shows positive ρ, and weak-signal environments (Plancraft) cluster near zero.

Backbone-level reversal is a prediction, not an anomaly. A subtlety is essential here: p_I is not a property of the environment alone. A state is information-poor only relative to what the agent already encodes; the same observation can be Type I for one model and Type D for another. The model backbone therefore enters the mixture directly, and the effective p_I is indexed by the (environment, backbone) pair. Read this way, the backbone-level sign flips of Table 1—most extreme on FEVER between Phi-3.5 and Llama-3.1—are exactly what the model predicts: backbone changes shift p_I, and when they cross p_I^*, ρ flips. Far from contradicting the two-source view, backbone reversal becomes one of its predictions (Appendix G.3). ======= negligible signal. This expression captures the sign-level dependence of the marginal trend on the mixture proportion under a simplified linear model; it is not intended as a precise estimator of Spearman ρ. The prediction is consistent with the observed Qwen3 pattern in Table 1: search-based QA (FEVER, HotpotQA, TWExpress) shows negative ρ, code generation (APPS) shows positive ρ, and weak-signal

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Theoretical grounding. This direction reversal is a special case of Simpson's paradox (Simpson, 1951; Pearl, 2022): aggregating subpopulations with opposing within-group trends reverses the aggregate correlation. Our decomposition adapts the epistemic/aleatoric distinction (Der Kiureghian and Ditlevsen, 2009) to the meta-decision setting: Type I states exhibit epistemic-like uncertainty (information deficit), while Type D states exhibit aleatoric-like diversity (multiple viable paths). We treat this two-source decomposition as a *minimal falsifiable* model rather than a post-hoc explanation: in Section 5.4 we derive three predictions from it (temporal dynamics, cross-environment consistency, and signal identity alignment) and verify each on held-out data.

¡¡¡¡¡¡¡ HEAD The decomposition also makes the insufficiency claim of §3.1 precise. Since the latent type $\tau(s) \in \{I, D\}$ governs the sign of $\mathrm{VOC}(s)$ but is not measurable from $\sigma(s)$ alone, no function of σ can recover τ, and therefore no function of σ can recover the sign of VOC. The information-theoretic content of the reversal is exactly this non-identifiability of τ from σ.

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Implication. Under this decomposition, σ is not a sufficient statistic for VOC: two states with identical σ but opposite latent types have opposite-sign utility. No policy of the form $g(\sigma(s))$ can therefore deliver non-negative value across heterogeneous settings, and uncertainty-only fixed-direction gates empirically fail across our six environments (Proposition 1, proof in Appendix B.1). Recovering sufficiency requires augmenting σ with features that distinguish the two state types, directly motivating the multi-feature design of §4.

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4 Direction-Informed Adaptive Learning

¡¡¡¡¡¡¡ HEAD Section 3 establishes that the uncertainty signal σ alone is not a sufficient statistic for the value of computation, so any cross-setting adaptive gate must condition on auxiliary features $\phi(s)$ that carry information about the latent state type $\tau(s)$. Concretely, this implies three method constraints: per-(environment, backbone) direction discovery (§3.1), a multi-signal feature pool that approximates τ (§3.1), and *signal-agnostic* exploration—developed below—since conditioning data collection on the very signal whose direction is unknown introduces a self-confirming bias.

DIAL is the minimal procedure that lifts σ toward sufficiency for VOC under these constraints (Fig. 3): randomized exploration (§4.1) collects an unbiased (s, U) dataset, an optional LLM feature layer (§4.2) extends the universal feature pool when needed, and a sparse linear gate (§4.3) selects which features carry information about τ in each (environment, backbone) setting. The sophistication of the classifier is not the contribution; the contribution is the demonstration that once σ is augmented with the right auxiliary features, the simplest possible classifier already outperforms every fixed-direction baseline (§5). ======= The analysis in Section 3 implies three constraints for any cross-setting adaptive gate: it must discover the signal–utility direction per (environment, backbone) pair, it must draw from a multi-signal feature pool that approximates the latent state type $\tau(s)$, and it must collect training data without conditioning on the very signal whose direction is unknown. DIAL satisfies these constraints through three stages (Fig. 3): randomized exploration (§4.1), feature construction (§4.2), and sparse gate fitting (§4.3). ¿¿¿¿¿¿¿ 1bf5b0e2c05afb339eb3e76f610bbf4ff034c5a3

4.1 Explore: Signal-Agnostic Data Collection

¡¡¡¡¡¡¡ HEAD Exploration collects the (s, U) pairs that will be used to fit the gate. The central constraint is signal-agnosticity: the collection policy must not condition on any signal whose semantics we are trying to learn. Although

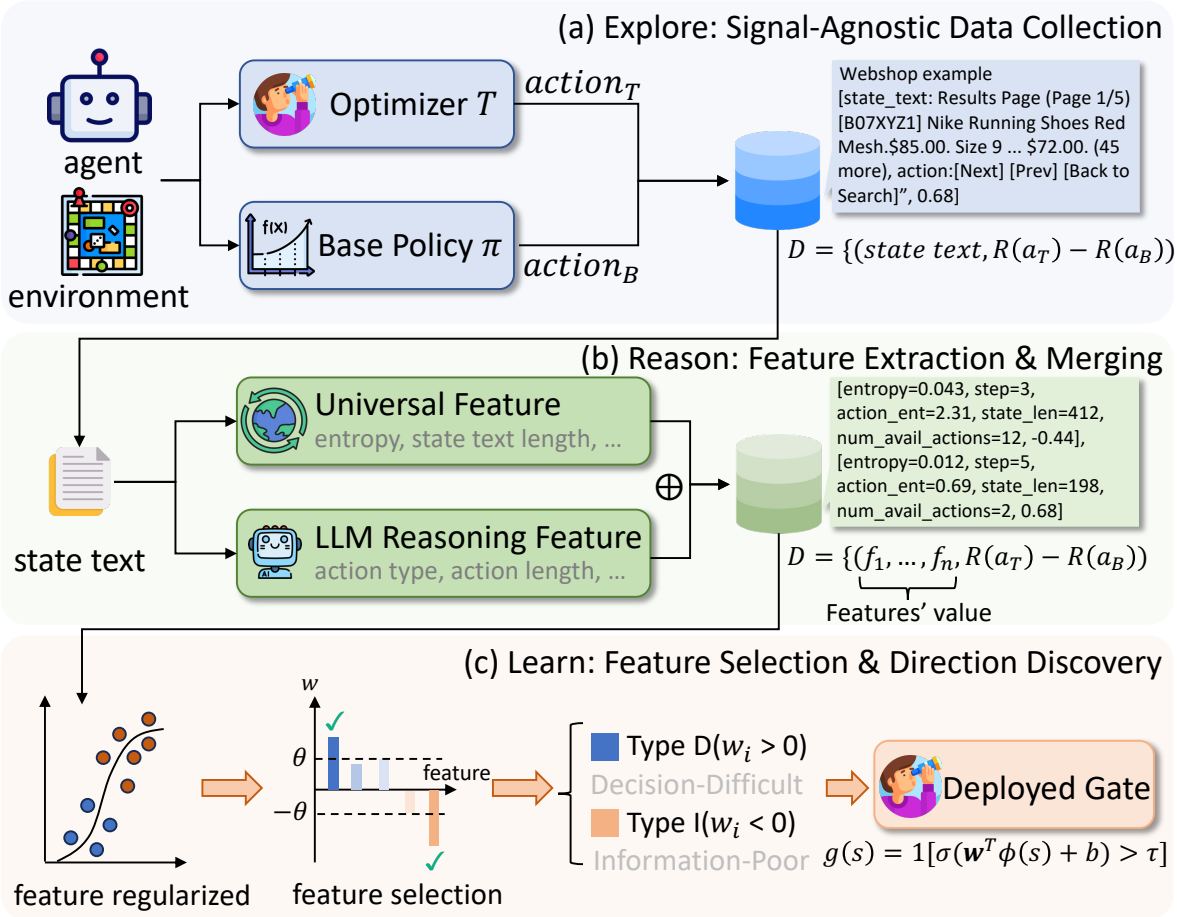

Figure 3: **The DIAL pipeline.** **(a) Explore**: collect signal-agnostic (s, U) pairs via Bernoulli-ε rollouts. **(b) Reason**: build a candidate feature pool $\phi(s)$ from universal metrics plus optional LLM-proposed task-specific features. **(c) Learn**: fit an ℓ_1-regularized logistic gate, whose signed weights jointly perform feature selection and recover per-environment direction (Type D: w_i>0; Type I: w_i<0).

uncertainty-conditioned data collection is the default in prior adaptive-gating work, it is incompatible with the goal of *learning* direction: a gate trained on uncertainty-triggered states systematically under-samples exactly the states that determine ρ's sign, biasing the fitted direction toward the prior assumption rather than testing it. DIAL therefore uses the simplest policy consistent with signal-agnosticity. At each step t in episode i, a uniform random coin flip ======= Exploration collects the (s, U) pairs used to fit the gate. The central constraint is signal-agnosticity: the collection policy must not condition on any signal whose semantics we are trying to learn. A gate trained on uncertainty-triggered states systematically under-samples exactly the states that determine ρ's sign, biasing the fitted direction toward the prior assumption rather than testing it. DIAL therefore uses the simplest policy consistent with signal-agnosticity. At each step t in episode i, a uniform random coin flip ¿¿¿¿¿¿¿ 1bf5b0e2c05afb339eb3e76f610bbf4ff034c5a3 $z_t^{(i)} \sim \text{Bernoulli}(\varepsilon_{\text{explore}})$ determines whether to trigger the optimizer T (z_t=1) or use the base action (z_t=0). Aggregating across N_{explore} episodes yields an approximately unbiased dataset (the Bernoulli trigger eliminates selection bias; the remaining approximation is the truncated-rollout estimator of U):

$$\mathcal{D} = \left\{ \left(\phi(s_t^{(i)}),\ U_t^{(i)} \right) : z_t^{(i)} = 1 \right\}_{t,i}, \qquad (3)$$

where $U_t^{(i)} = \mathbf{1}[V(a_T) > V(a_B)]$ indicates whether the optimizer's selected action outperformed the base action, as estimated by paired counterfactual rollouts from the same state snapshot, and $\phi(s)$ is the feature vector described in §4.2.

Estimating U. For each explored state with z_t=1, we estimate utility by counterfactual evaluation from the same state snapshot: we fork the

environment, execute both the optimizer-selected action a_T and the base action a_B via truncated rollouts of horizon H, and set $U_t = \mathbf{1}[V(a_T) > V(a_B)]$. This paired design eliminates selection bias; the remaining approximation is the truncation horizon. Per-environment estimation details are in Appendix D.1.

4.2 Reason: LLM Feature Layer

The next step builds the feature pool $\phi_{\text{cand}}(s)$ that the gate will be fit over. This pool must be rich enough to include each environment's relevant signals, but not so environment-specific that deployment requires manual engineering per task.

DIAL's default is a fixed set of universal features: `step_count`, `token_entropy`, `state_length`, `action_entropy`, and others. These are available in every environment without task-specific engineering. In the environments we study, this pool alone is sufficient to recover a gate that outperforms all fixed-direction baselines (§5.3).

For environments where universal features may be inadequate, ¡¡¡¡¡¡¡ HEAD such as WebShop, whose gating benefits from task-specific signals like `num_available_actions`, DIAL offers an *optional* automation layer: an LLM is given a summary of $\mathcal{D}$ and proposes task-specific feature hypotheses $\phi_{\text{LLM}}(s)$, which join the universal pool to form $\phi_{\text{cand}}(s) = \phi_{\text{univ}}(s) \cup \phi_{\text{LLM}}(s)$. Details of the LLM input format and prompt template are in Appendix C.3. The ablation in §5.3 shows that this layer is load-bearing only on FEVER (+9.1 pp); on every other environment its contribution is $\leq$1.8 pp, and on APPS and Plancraft it is effectively zero. We therefore present the LLM layer as an honest extensibility mechanism rather than a core component, and report all main-result numbers without claiming its contribution is general. ======= DIAL offers an optional automation layer: an LLM is given a summary of $\mathcal{D}$ and proposes task-specific feature hypotheses $\phi_{\text{LLM}}(s)$, which join the universal pool to form $\phi_{\text{cand}}(s) = \phi_{\text{univ}}(s) \cup \phi_{\text{LLM}}(s)$. The ℓ_1 penalty in the gate fitting stage (§4.3) automatically discards uninformative proposals. Details of the LLM input format and prompt template are in Appendix C.3; the ablation in §5.3 quantifies the contribution of this layer per environment. ¿¿¿¿¿¿¿ 1bf5b0e2c05afb339eb3e76f610bbf4ff034c5a3

4.3 Learn: Sparse Linear Gate

Given $\mathcal{D}$ and ϕ_{cand}, DIAL fits an ℓ_1-regularized logistic regression:

$$\mathbf{w}^*, b^* = \arg\min_{\mathbf{w},b} \sum_{(\phi,U)\in\mathcal{D}} \ell\big(\sigma(\mathbf{w}^\top\phi + b),\, U\big) + \lambda\|\mathbf{w}\|_1 \tag{4}$$

where ℓ is the binary cross-entropy loss and λ controls sparsity over ϕ_{cand} so that uninformative features drop out. The deployed gate is

$$g(s) = \mathbf{1}\Big[\sigma({\mathbf{w}^*}^\top\phi(s) + b^*) > \tau\Big], \tag{5}$$

with threshold τ set by cross-validation on $\mathcal{D}$. ¡¡¡¡¡¡¡ HEAD

Cost accounting. DIAL incurs three costs: (i) a one-time exploration cost of N_{explore}=50 episodes per (environment, backbone) pair, amortized over all subsequent deployment; (ii) sub-second offline gate-fitting time, dominated by feature extraction; and (iii) per-step inference cost equal to evaluating $\phi(s)$ and a single sigmoid. Only (iii) is incurred at deployment, and it is zero relative to a non-gated agent: there is no extra LLM call, no K-way voting, and no calibration query. Implementation details are in Appendix C. ======= At deployment, the gate requires only a feature evaluation and a single sigmoid, adding no extra LLM calls. ¿¿¿¿¿¿¿ 1bf5b0e2c05afb339eb3e76f610bbf4ff034c5a3

Signed weights as a diagnostic. The ℓ_1 penalty in Eq. 4 simultaneously performs feature selection and direction discovery. The sign of each fitted weight provides an interpretable per-environment estimate of the Two-Source mixture of §3.2: $w_i > 0$ indicates the environment sits on the Type D side of the reversal threshold p_I^* for signal i (high signal $\rightarrow$ trigger), $w_i < 0$ the Type I side (high signal $\rightarrow$ *don't* trigger), and $w_i \approx 0$ an uninformative signal. The complete pipeline, including online adaptation for drifting environments and detailed cost accounting, is provided in Appendix C.

5 Experiments

5.1 Experimental Setup

We evaluate DIAL along three axes: does environment-aware gating Pareto-dominate fixed-direction baselines (§5.2); which components drive the gains (§5.3); and do the Two-Source Model's predictions hold on held-out data (§5.4).

Environments. We evaluate on 6 environments: HotpotQA (Yang et al., 2018), APPS (Hendrycks et al., 2021), WebShop (Yao et al., 2022), FEVER (Thorne et al., 2018), TWExpress (Jansen and Côté, 2023), and Plancraft (Dagan et al., 2024). These cover the full range of the Two-Source Model, from pure Type I (FEVER) through mixed (WebShop) to Type D (APPS), including extreme rollout properties (TWExpress: rollout-safe; Plancraft: rollout-harmful). Full environment specifications are in Appendix D.1.

Baselines. All methods share the same agent and optimizer T; only the gating decision differs. We include two bounds: *base_only*, which never triggers the optimizer, and *always_trigger*. Six fixed-direction baselines all assume that high uncertainty warrants additional compute: CaTS (Huang et al., 2026a), SEAG (Lee et al., 2025), CoRefine (Jin et al., 2026), CATTS (Lee et al., 2026), AUQ (Zhang et al., 2026), and s1_budget (Muennighoff et al., 2025). Full specifications are in Appendix D.2.

Metrics. SR (success rate, ↑) and Cost (×base, ↓): total deployment tokens normalized by base_only tokens per episode. This metric counts *all* LLM calls, including base proposer, gate overhead (CATTS K=5 voting, AUQ confidence queries), and rollout calls, ensuring that per-step overhead is not hidden. Calibration or exploration overhead is excluded for all methods. A method Pareto-dominates another if SR $\geq$ and Cost $\leq$ with at least one strict inequality. Token cost constants are reported in Appendix D.3.

5.2 Main Results

DIAL Pareto-dominates fixed-direction baselines. Figure 4 shows the SR–Cost Pareto frontier across all 6 environments on Qwen3-4B. Full results across three backbones are in Appendix E.

Three patterns emerge. First, *fixed-direction baselines catastrophically fail when the assumed direction is wrong*: on FEVER (Type I), CATTS achieves only 32.2% SR (below the 37.0% no-trigger baseline) yet costs 16.46×base due to its K=5 per-step voting overhead. Second, *DIAL achieves competitive SR at substantially lower cost*. On WebShop, DIAL reaches 43.8% SR at 2.50×base, matching always_trigger (43.0%) at less than half the cost (5.56×). On APPS, DIAL outperforms the best baseline (s1_budget, 69.0%) by 4 pp at comparable cost. Third, *in rollout-harmful environments, DIAL learns to suppress triggering*: in Plancraft (Δ=−7.0), DIAL recovers near base_only performance (23.3% vs. 29.8%), while most baselines that trigger aggressively fall below base_only (s1_budget: 16.3%).

Table 2: DIAL vs. bounds on Qwen3-4B. Δ: rollout headroom (always−base).

	base_only		**always_trigger**		**DIAL**	
Env	SR↑	Cost↓	SR↑	Cost↓	SR↑	Cost↓
HotpotQA	49.0	1.00	97.0	10.63	95.2	8.02
WebShop	7.2	1.00	43.0	5.56	43.8	2.50
FEVER	37.0	1.00	59.8	17.87	49.8	16.51
TWExpress	67.5	1.00	99.3	2.15	99.0	1.81
Plancraft	29.8	1.00	22.8	6.13	23.3	3.63
APPS	60.5	1.00	79.5	3.65	73.0	2.61

DIAL adapts its behavior to each environment. Table 2 compares DIAL against the two bounds on Qwen3-4B. In environments with positive rollout headroom (HotpotQA, WebShop, TWExpress, APPS), DIAL achieves SR close to always_trigger at substantially lower cost. In the rollout-harmful environment Plancraft (Δ=−7.0), DIAL learns to suppress triggering, losing only 6.5 pp relative to base_only while always_trigger loses 7.0 pp. FEVER remains the weakest case: DIAL improves over base_only (49.8% vs. 37.0%) but falls far short of always_trigger (59.8%), a limitation analyzed in §6.

5.3 Ablation Studies

The main results show *that* DIAL works; we now ask *why*. We ablate the two key components: direction learning and the LLM feature layer.

Direction is a first-order determinant of gate quality. To isolate the effect of direction, we reverse the sign of all learned weights in DIAL's fitted gate, keeping everything else unchanged. Table 3 shows the result. In environments with strong signal ($|\rho| > 0.4$), reversal causes SR drops exceeding 23 pp. In weak-signal environments like Plancraft ($|\rho|$=0.016), reversal has negligible effect. The degradation scales with signal strength: wrong direction is not a minor calibration error but a structural failure, consistent with the wrong-direction harm result of ¡¡¡¡¡¡¡ HEAD Appendix B.1. *This table is the direct empirical statement of the headline claim*: the gate whose weights are reversed is, by construction, a more precisely calibrated harm-targeting mechanism, and its 23–37 pp SR collapse on strong-signal environments is the price of that precision when

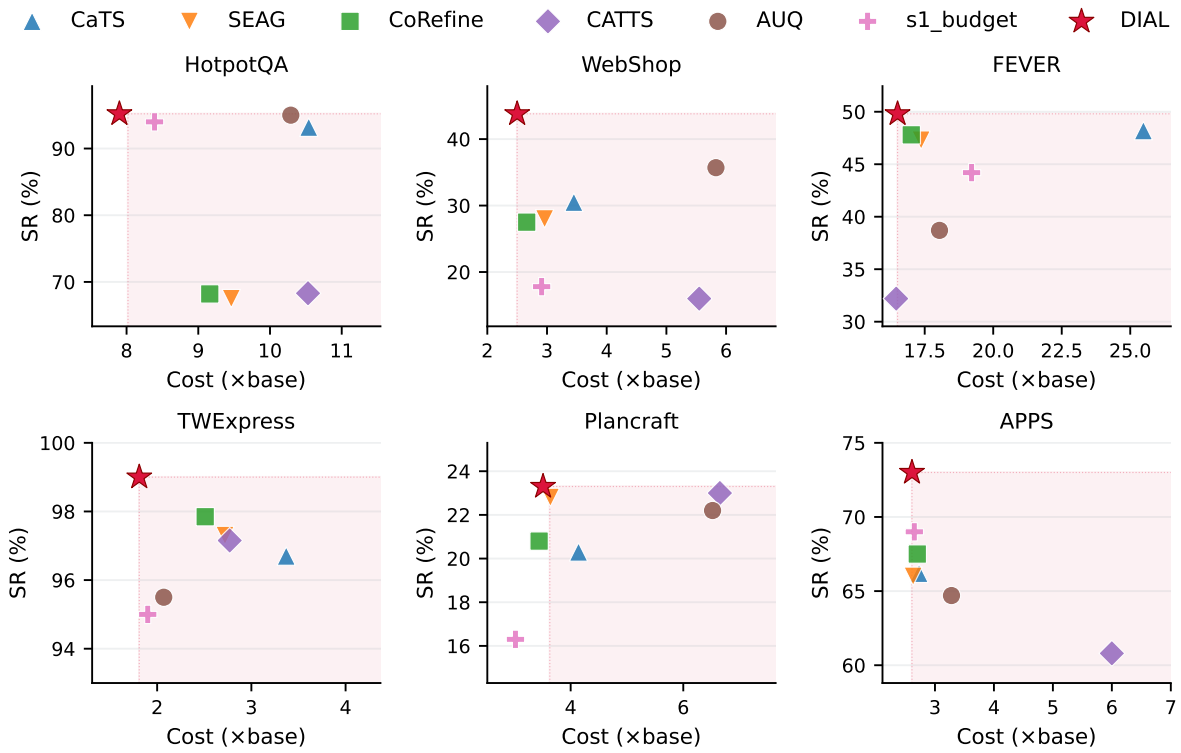

Figure 4: SR vs. Cost (×base) across all 6 environments (Qwen3-4B). Shaded region: dominated by DIAL (lower SR *and* higher cost). Full numerical results across three backbones are in Table 10.

Table 3: Effect of reversing DIAL's learned direction. ΔSR: reversed − original. Degradation scales with signal strength $|\rho|$.

Environment	DIAL SR	Reversed SR	Δ SR	$\lvert\rho\rvert$
FEVER	49.8%	13.0%	−36.8	0.619
HotpotQA	95.2%	58.2%	−37.0	0.494
WebShop	43.8%	20.6%	−23.2	0.444
APPS	73.0%	70.5%	−2.5	0.339
TWExpress	99.0%	96.2%	−2.8	0.290
Plancraft	23.3%	20.5%	−2.8	0.016

Table 4: LLM feature ablation. SR (%) with and without LLM-generated features. FEVER is the only environment with meaningful LLM contribution.

Environment	w/ LLM	w/o LLM	Δ
HotpotQA	95.2	93.8	+1.4
WebShop	43.8	42.7	+1.1
FEVER	49.8	40.7	+9.1
TWExpress	99.0	97.2	+1.8
Plancraft	23.3	23.4	−0.2
APPS	73.0	73.0	+0.0

the direction is wrong. ======= Appendix B.1. Reversing the learned weights constructs an adversarially wrong-direction gate: it preserves the learned ranking structure while inverting the trigger semantics. The resulting 23–37% SR collapse on strong-signal environments confirms that ranking states well is harmful when the direction is inverted, as the gate fires precisely on states where computation has negative value. ¿¿¿¿¿¿¿ 1bf5b0e2c05afb339eb3e76f610bbf4ff034c5a3

The LLM feature layer is an automation convenience. Table 4 compares DIAL with and without LLM-generated features. FEVER is the only environment where LLM features contribute meaningfully (+9.1 pp), where task-specific signals (`has_claim`, `text_length`) extend the universal feature pool. In the remaining environments, removing LLM features changes SR by <2 pp; in Plancraft and APPS the contribution is effectively zero.

ℓ_1 is not interchangeable with other shrinkage or selection schemes. The signed-weight diagnostic of §4.3 and DIAL's tolerance of noisy LLM-proposed features both rely on the *exact* sparsity induced by ℓ_1. Replacing it with ℓ_2 yields comparable SR on strong-signal environments but eliminates the clean {informative, uninformative} partition, because every feature retains a small non-zero weight; unregularized lo-

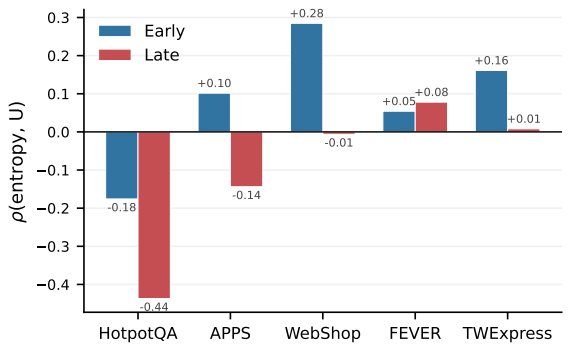

Figure 5: P1 verification: ρ(entropy, U) shifts from early to late steps. Blue: early (step $\leq$ median), red: late. Values on each bar show the Spearman ρ for that phase. Plancraft is omitted ($|\rho|$=0.016, too weak for meaningful comparison).

gistic overfits on weak-signal environments like Plancraft; hard mutual-information feature selection recovers SR on strong-signal tasks but requires per-environment tuning of k, reintroducing the engineering cost DIAL aims to eliminate. The full comparison across regularizers, holding $\mathcal{D}$ and ϕ_{cand} fixed, is reported in §G.5.

5.4 Two-Source Model Verification

The ablations confirm DIAL's practical advantage; we now test whether the Two-Source Model (§3.2) has predictive power beyond explaining the observed correlations in Table 1. Following Schaeffer et al. (2023), we derive three testable predictions and verify each.

P1: Temporal dynamics. The Two-Source Model predicts that ρ should decrease over the episode: as the agent gathers information, Type D states are resolved and late-step uncertainty increasingly reflects residual Type I states. Figure 5 confirms this. In HotpotQA, ρ shifts from -0.176 (early) to -0.437 (late). In WebShop, the strong positive early-phase signal (ρ=$+0.285$) vanishes in the late phase. In APPS, ρ reverses from positive ($+0.102$) to negative (-0.144). Full numerical results are in Appendix G.1.

P2: Cross-setting consistency. The Two-Source Model predicts that settings with similar p_I should exhibit similar ρ. With the backbone held fixed at Qwen3 (Table 1), this holds: FEVER and HotpotQA (both search-based QA) share negative ρ; WebShop and APPS (both involving choice among alternatives) share positive ρ. As predicted by the (environment, backbone)-indexed mixture of §3.2, however, task-structure similarity alone does not pin down ρ when the backbone changes: the same environment can sit on opposite sides of p_I^* under different LLMs (Appendix G.3).

Table 5: Controlled information manipulation on HotpotQA. Manipulating information availability shifts which signal dominates, confirming the Two-Source Model's prediction.

Condition	Base SR	ρ(entropy)	ρ(step_count)	DIAL SR	n
Original	49.0%	-0.041	-0.494	95.2%	1117
InfoPoor	44.0%	$+0.119$	-0.608	94.0%	317
InfoRich	82.0%	$+0.311$	-0.147	88.3%	221

P3: Signal identity alignment. The Two-Source Model predicts that the strongest signal should reflect the dominant state type: information-sufficiency proxies in Type I environments, decision-complexity proxies in Type D environments. Appendix G.2 confirms this: `step_count` dominates in Type I environments, while `num_available_actions` dominates in WebShop.

To provide causal evidence, we manipulate information availability within HotpotQA itself. Table 5 shows the results: restricting evidence (InfoPoor) makes `step_count` the dominant signal (ρ=-0.608) while entropy is weak; injecting gold evidence (InfoRich) makes entropy dominant (ρ=$+0.311$) while `step_count` weakens. This confirms that signal identity is a function of information structure, not an environment-specific constant. DIAL adapts to both conditions without manual reconfiguration.

All three predictions are supported. Additional robustness checks are in Appendix G: multi-backbone verification (§G.3), stratified analysis (§G.4), and gate complexity ablations (§G.6).

6 Limitations

DIAL's weakest result is on FEVER, where all methods incur high cost due to the environment's extreme Type I structure: DIAL costs 16.51×base and achieves 49.8% SR, while cheaper alternatives like SEAG (47.3% at 17.19×) are only marginally lower in both SR and cost. Rollout value in FEVER concentrates in the first one or two steps, and random exploration misses these critical steps with 50% probability. This failure mode is predicted by the two-source model: FEVER is the most extreme Type I environment, where undirected exploration is least efficient, and

exposes the cost of insisting on signal-agnostic exploration in high-p_I regimes. Addressing this through curriculum-based exploration is an important direction for future work. We verify our findings on three backbones (4B–8B) across all six environments (Appendix G.3); testing on larger models (70B+) remains future work. The two-source model assumes linear conditional relationships and serves as an explanatory framework rather than a precise predictor of ρ values. Finally, DIAL requires N_{explore}=50 exploration episodes per new (environment, backbone) deployment—a modest but non-zero one-time cost amortized over subsequent inference, and the only cost incurred beyond the per-step inference budget reported in §4.

7 Conclusion

¡¡¡¡¡¡¡ HEAD The central claim of this paper is that *uncertainty signals are not sufficient statistics for the value of computation*. Existing adaptive gating methods for LLM agents share an implicit fixed-direction assumption—higher uncertainty indicates when extra computation helps—which is the strict-monotone special case of a structurally insufficient family: any policy depending on the uncertainty signal alone, no matter how it is calibrated or post-processed, must underperform never-gating on at least one (environment, backbone) setting. The reason is that the value of computation depends on a latent state property—information poverty versus decision difficulty—that the signal alone cannot identify and whose mixture is determined jointly by the environment and the backbone. Across six environments and three model backbones, signal–utility direction reverses along both axes, and once misaligned, better calibration *worsens* performance by more precisely selecting harmful states. A two-source model grounded in Simpson's paradox explains when and why reversal occurs, makes a falsifiable prediction we verify on held-out data, and identifies what an adaptive-compute method must condition on to recover sufficiency: features that distinguish the two state types. As an existence proof that this auxiliary conditioning—not gate sophistication—is the bottleneck, DIAL augments the uncertainty signal with a small pool of universal state features and fits a sparse linear gate on signal-agnostic exploration data, adding zero per-step inference overhead. Across every setting evaluated, this minimal procedure already dominates fixed-direction baselines, and its full range of gating behavior emerges from direction learning alone. ======= Existing adaptive gating methods for LLM agents assume that higher uncertainty consistently indicates when extra computation helps. We show this fixed-direction assumption fails: the signal–utility relationship reverses across both environments and model backbones, and when the assumed direction is wrong, better calibration worsens performance by more precisely selecting harmful states. We trace this reversal to a mixture of information-poor and decision-difficult states whose composition varies across (environment, backbone) pairs, and show that, under the two-source model, no policy depending on uncertainty alone can guarantee non-negative value across heterogeneous settings. DIAL learns the correct direction per setting by fitting a sparse linear gate over a small set of state features, and outperforms every fixed-direction baseline across six environments and three backbones. ¡¡¡¡¡¡¡ HEAD ======= ¿¿¿¿¿¿¿ 1bf5b0e2c05afb339eb3e76f610bbf4ff034c5a3 ¿¿¿¿¿¿¿ 3da75c3f8977898076ea83cf9ca97c6d089732a5

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A Extended Related Work

A.1 Adaptive Compute: Reasoning vs. Agent Settings

The adaptive test-time compute literature spans two distinct settings. In the *reasoning* setting, methods optimize single-turn chain-of-thought generation on homogeneous benchmarks (MATH, GSM8K, GPQA). Signal-based approaches use token-level confidence or entropy to decide when to extend thinking (Lee et al., 2025; Huang et al., 2026a; Jin et al., 2026; Sharma and Chopra, 2025; Muennighoff et al., 2025; Han et al., 2025). RL-based methods learn think/no-think policies through reinforcement learning (Zhang et al., 2025; Fang et al., 2025; Aggarwal and Welleck, 2025). Hidden-state probing methods train lightweight classifiers on internal representations to estimate task difficulty (Liu et al., 2025; Zhu et al., 2025b), sometimes assuming a universal U-shaped entropy pattern that our cross-(environment, backbone) evidence disproves. Because reasoning benchmarks share similar task semantics, signal direction happens to be consistent across tasks, and the fixed-direction assumption is never tested.

In the *agent* setting, methods operate in multi-step interactive environments with external optimizers. Vote-based methods (Lee et al., 2026; Jin et al., 2026) sample multiple candidate actions and trigger evaluation when disagreement is high. Verbalized-uncertainty methods (Zhang et al., 2026) use self-reported confidence for agent reflection but assume fixed direction. Episode-level approaches (Paglieri et al., 2025) learn when to invoke a planner, and agentic RL methods (Dong et al., 2025) use entropy-based adaptive rollout at tool-call steps. All agent-setting methods inherit the fixed-direction assumption from the reasoning literature without verifying it. Our work is the first to systematically study signal semantics across heterogeneous agent environments *and backbones*, where we find that the direction reverses along both axes.

A.2 Adaptive Computation in Neural Networks

The idea of adapting computation to input difficulty predates LLMs. Adaptive Computation Time (Graves, 2016) allows RNNs to learn how many computational steps to take per input. Early exit methods (Teerapittayanon et al., 2017; El-

houshi et al., 2024) attach classifiers at intermediate transformer layers and exit early for easy inputs. Mixture-of-Experts architectures (Shazeer et al., 2017; Raposo et al., 2024) activate only a subset of parameters per token, adapting capacity rather than depth. These methods adapt computation within a single forward pass. Our work operates at a different granularity: we adapt whether to invoke an external optimizer at each decision step, and the key challenge is not input difficulty but the *direction* of the signal–utility relationship.

A.3 LLM Confidence Estimation and Calibration

Kadavath et al. (2022) show that larger models are well-calibrated on multiple-choice tasks and can estimate P(True) for their own outputs. Tao et al. (2025) and Heo et al. (2025) find that calibration quality varies substantially across task types, providing converging evidence that uncertainty semantics are domain-dependent. Abbasi Yadkori et al. (2024) propose decoupling epistemic and aleatoric uncertainty in LLM outputs, a distinction conceptually related to our Type I/Type D decomposition. Our work extends these findings: not only does calibration *quality* vary across tasks, but the *direction* of the uncertainty–utility relationship reverses entirely.

A.4 Metareasoning and Value of Computation

Rational metareasoning (Russell and Wefald, 1991; Horvitz, 1987) formalizes the value of computation (VOC) as a guide for allocating cognitive resources. The classical framework assumes VOC ≥ 0 because an agent can always ignore computation results. Lieder and Griffiths (2020) generalize this to resource-rational analysis, modeling cognition as optimal use of limited computational resources. Sabbata et al. (2024) apply metareasoning to LLMs, analyzing when to extend chain-of-thought reasoning. Our work shows that the classical VOC ≥ 0 assumption is violated in the agent setting; we formalize this in Appendix B.3.

A.5 Simpson's Paradox in Machine Learning

Simpson's paradox (Simpson, 1951; Pearl, 2022) occurs when a trend present in subgroups reverses upon aggregation. In machine learning, this phenomenon has been documented in fairness (Wang et al., 2023a), causal inference, and treatment effect estimation. Our two-source model identifies a new instance: within Type I states, entropy negatively predicts utility; within Type D states, entropy positively predicts utility; but the marginal correlation depends on the mixture proportion p_I, which is itself an (environment, backbone)-indexed quantity, since a state's status as Type I or Type D depends on what the agent already encodes, so the same environment can sit on opposite sides of the reversal threshold p_I^* under different LLMs. This makes direction reversal a structural consequence of aggregating heterogeneous uncertainty sources, and predicts both the across-environment and across-backbone reversals reported in Table 1.

A.6 Search-Based Optimizers

A growing body of work targets *what* the test-time compute procedure does, rather than *when* it should be invoked. Tree-search and lookahead methods organize multi-step exploration through value propagation and AlphaZero-style rollouts (Yao et al., 2023a; Zhou et al., 2024; Hao et al., 2023; Wang et al., 2026; Wan et al., 2024; Misaki et al., 2025). Repeated independent sampling with majority aggregation exploits smooth coverage gains as the sample budget grows (Wang et al., 2023b; Brown et al., 2024), while verifiers trained over intermediate reasoning steps rerank candidates or steer search (Lightman et al., 2024; Wang et al., 2024). At a higher level, compute-allocation analyses and reasoning models study how a fixed budget should be distributed across these mechanisms or absorbed into long latent traces (Snell et al., 2024; DeepSeek-AI, 2025). Closer to our setting, agent-specific work adapts these mechanisms to interactive, multi-step environments by applying tree search to web tasks (Koh et al., 2024), coupling MCTS with preference learning over agent trajectories (Putta et al., 2024), and characterizing scaling behavior for agent rollouts (Zhu et al., 2025a; Huang et al., 2026b). All of this work specifies the rollout procedure that runs at a triggered step. Our contribution is orthogonal: a per-step gate decides *whether* such a procedure should run, and the gate composes with any of the above as a drop-in test-time optimizer.

B Theoretical Material

This appendix collects the theoretical material referenced in Section 3: the formal statement and proof of the necessity-of-direction-discovery result, the wrong-direction damage quantification, and the VOC negativity analysis.

B.1 Insufficiency of Uncertainty Signals: Statement and Proof

We formalize the central theoretical claim of the paper: any adaptive-compute policy that depends on an uncertainty signal σ alone cannot deliver non-negative value of computation across heterogeneous (environment, backbone) settings.

Proposition 1 (Insufficiency of Uncertainty Signals)**.** *Let $\sigma : \mathcal{S} \to \mathbb{R}$ be any uncertainty signal (e.g., token entropy, calibrated confidence, vote disagreement, verbalized uncertainty). Let $\mathrm{VOC}(s) = \mathbb{E}[R(\tau) \mid a{=}T(s)] - \mathbb{E}[R(\tau) \mid a{=}\pi(s)]$ denote the value of computation at state s. Then σ is* not a sufficient statistic *for* VOC *in the following sense: there exist (environment, backbone) settings $\mathcal{E}_1, \mathcal{E}_2$ and states $s_1 \in \mathcal{E}_1$, $s_2 \in \mathcal{E}_2$ with $\sigma(s_1) = \sigma(s_2)$ such that $\mathrm{VOC}(s_1) > 0$ and $\mathrm{VOC}(s_2) < 0$. Consequently, no policy of the form $\pi_\sigma(s) = g(\sigma(s))$ for any measurable $g : \mathbb{R} \to \{0, 1\}$ can simultaneously satisfy $\mathrm{SR}(\pi_\sigma, \mathcal{E}_1) \geq \mathrm{SR}(\mathrm{base}, \mathcal{E}_1)$ and $\mathrm{SR}(\pi_\sigma, \mathcal{E}_2) \geq \mathrm{SR}(\mathrm{base}, \mathcal{E}_2)$.*

The proposition strengthens the earlier "no fixed-direction gate works" formulation in two ways. First, it rules out *any* measurable $g(\sigma)$, not just monotone or threshold-based policies, so non-monotone or U-shaped calibrators are also covered. Second, it identifies the underlying obstruction as an information-theoretic property of σ itself, one that survives any post-processing of the signal.

Proof. By the two-source decomposition of §3.2, $\mathrm{VOC}(s)$ depends on a latent type $\tau(s) \in \{I, D\}$ that is not measurable from $\sigma(s)$ alone:

$$\mathrm{VOC}(s) = \begin{cases} -\alpha\,\sigma(s) + \varepsilon_I & \text{if } \tau(s) = I, \\ +\beta\,\sigma(s) + \varepsilon_D & \text{if } \tau(s) = D, \end{cases}$$

with $\alpha, \beta > 0$. Choose any signal value σ_0 in the support of both Type I and Type D states within their respective settings. Then there exist $s_1 \in \mathcal{E}_1$ with $\tau(s_1) = D$ and $\sigma(s_1) = \sigma_0$, giving $\mathbb{E}[\mathrm{VOC}(s_1) \mid \sigma_0] = \beta\sigma_0 > 0$ for sufficiently large σ_0, and $s_2 \in \mathcal{E}_2$ with $\tau(s_2) = I$ and $\sigma(s_2) = \sigma_0$, giving $\mathbb{E}[\mathrm{VOC}(s_2) \mid \sigma_0] = -\alpha\sigma_0 < 0$. Hence the conditional VOC is not a function of σ alone, establishing insufficiency.

For the policy claim, fix any $g : \mathbb{R} \to \{0, 1\}$. The expected one-step reward gain over base from playing π_σ is $\mathbb{E}_{s\sim\mathcal{E}}[g(\sigma(s)) \cdot \mathrm{VOC}(s)]$. Decomposing by latent type and writing $p_I(\mathcal{E})$ for the Type I fraction,

$$\begin{aligned}\mathbb{E}_{\mathcal{E}}[g(\sigma) \cdot \mathrm{VOC}] &= -\alpha\, p_I(\mathcal{E})\, \mathbb{E}_I[g(\sigma)\sigma] \\ &\quad + \beta\,(1 - p_I(\mathcal{E}))\, \mathbb{E}_D[g(\sigma)\sigma] + \mathrm{const}.\end{aligned}$$

Whenever g is non-trivially correlated with σ (which any informative gate must be), the two terms have opposite signs as p_I varies. In particular, for any fixed g there exist $\mathcal{E}_1, \mathcal{E}_2$ with $p_I(\mathcal{E}_1)$ and $p_I(\mathcal{E}_2)$ on opposite sides of the reversal threshold $p_I^* = \beta/(\alpha + \beta)$ of §3.2 such that the expected gain has opposite signs. The setting where it is negative violates the SR constraint relative to base, completing the argument. □

Corollary (necessity of direction discovery). The original "no fixed-direction gate works" result is the restriction of Proposition 1 to threshold policies $g(\sigma) = \mathbf{1}[d \cdot \sigma > \theta]$, $d \in \{+1, -1\}$. In that case both cases of the proof above collapse to: g's sign of correlation with σ is fixed at d, but VOC's sign of correlation with σ flips with p_I, so on the wrong side of p_I^* the gate triggers exactly when $\mathrm{VOC} < 0$ and incurs systematic harm.

What recovers sufficiency. The proposition pinpoints what an adaptive-compute method must condition on: any feature $\phi(s)$ that is informative about the latent type $\tau(s)$ recovers a sufficient statistic for VOC, since $\mathrm{VOC}(s) = f(\sigma(s), \tau(s))$ is by construction a function of (σ, ϕ). DIAL's multi-feature gate (§4) is the minimal procedure that operationalizes this recovery: it augments σ with auxiliary universal features that empirically correlate with τ (e.g., `step_count` as a proxy for information accumulation, `num_available_actions` as a proxy for decision optionality) and lets the data identify which of them carry information about τ in each (environment, backbone) setting.

B.2 Wrong-Direction Damage: The Calibration-Worsens Mechanism

Table 3 is the empirical statement of the headline claim: a gate with reversed weights is, by construction, a more precisely calibrated *harm-targeting* mechanism, and the resulting SR col-

lapse scales with signal strength $|\rho|$. The decomposition predicts the relationship quantitatively. By the wrong-direction lemma in App B.1,

$$\mathrm{SR}(g_{-d^*}, \mathcal{E}) - \mathrm{SR}(\mathrm{base}, \mathcal{E}) \approx -c \cdot |\rho(\mathcal{E})|,$$

where c is a constant that depends on the optimizer's harm magnitude. Empirically this scaling is clean: HotpotQA ($|\rho|$=0.494) loses 37.0 pp, FEVER ($|\rho|$=0.619) loses 36.8 pp, WebShop ($|\rho|$=0.444) loses 23.2 pp, while Plancraft ($|\rho|$=0.016) loses only 2.8 pp; the slope of damage in $|\rho|$ is the slope of calibration quality, since better-calibrated gates concentrate their firing on higher-$|\sigma|$ states which, when the direction is wrong, are exactly the harmful ones.

Below we discuss two environments where reversal has minimal effect, both consistent with the same prediction (small $|\rho|$ or rollout-safe optimizer).

TWExpress is rollout-safe (Δ=+31.8): the optimizer rarely harms performance regardless of when it is triggered, so both directions achieve near-perfect SR (99.0%).

APPS shows a small negative ΔSR (-2.5 pp), consistent with its unstable signal direction across backbones (Table 1): Qwen3 shows positive ρ, Llama-3.1 shows negative ρ, and Phi-3.5 is non-significant. When the signal is this unstable, direction reversal has minimal impact and small fluctuations can favor either direction.

B.3 VOC Non-Negativity Scope

In the classical VOC framework (Russell and Wefald, 1991), $\mathrm{VOC}(T, s) = \mathbb{E}[\max(V(a_T), V(a_{\mathrm{base}}))] - V(a_{\mathrm{base}}) \geq 0$ because the agent can *disregard* the computation result if unfavorable. Our setting violates this: when the agent uses an evaluator/optimizer T, the evaluator's assessment *is* the agent's decision. The agent cannot separately evaluate the evaluator's output because the evaluator *is* the evaluation mechanism.

Under evaluator-executor identity, VOC can be negative: $\mathrm{VOC}(T, s) = \mathbb{E}[V(a_T(s))] - V(a_{\mathrm{base}}(s))$. If T systematically selects worse actions in Type I states, $\mathrm{VOC} < 0$ for those states. This explains why wrong-direction gating is not merely wasteful ($\mathrm{VOC} = 0$) but actively harmful ($\mathrm{VOC} < 0$), and why *better-calibrated* gates fire more precisely on these $\mathrm{VOC} < 0$ states, so calibration improvements amplify rather than mitigate the harm (Table 3 and App B.2). The classical VOC framework's $\mathrm{VOC} \geq 0$ guarantee fails for adaptive compute in agent settings precisely because the agent cannot "disregard" the optimizer's output, and any policy that depends only on σ (which cannot identify Type I states) will trigger T on a non-trivial fraction of $\mathrm{VOC} < 0$ states.

C DIAL Implementation Details

This appendix expands on the implementation of the σ-augmentation procedure introduced in Section 4: how the (s, U) data is collected under the signal-agnosticity constraint (Phase 1, §C.1), how the candidate feature pool ϕ_{cand} that approximates the latent type τ is constructed (Phase 2, §C.3), and how the sparse linear gate that lifts σ toward sufficiency for VOC is fit (Phase 3, §C.1).

Algorithm 1 DIAL: Direction-Informed Adaptive Learning

Require: Environment $\mathcal{E}$, agent policy π, optimizer T, universal feature set ϕ_{univ}, LLM $\mathcal{M}$, exploration budget N_{explore}, exploration rate $\varepsilon_{\text{explore}}$, gate threshold τ
Ensure: Deployed gate $g : \mathcal{S} \to \{0, 1\}$
// Phase 1: Explore
$\mathcal{D} \leftarrow \emptyset$
for episode $= 1, \ldots, N_{\text{explore}}$ **do**
  Run π in $\mathcal{E}$; at each step t, invoke T with probability $\varepsilon_{\text{explore}}$ independent of state
  Record (s_t, U_t) pairs into $\mathcal{D}$
end for
// Phase 2: Reason
$(p_{\mathcal{E}}, \mathcal{H}) \leftarrow \mathcal{M}(\mathrm{summary}(\mathcal{D}))$ ▷ env profile + feature hypotheses
$\phi_{\text{cand}} \leftarrow \phi_{\text{univ}} \cup \mathrm{extract}(\mathcal{H})$
Recompute $\phi_{\text{cand}}(s)$ for all $s \in \mathcal{D}$
// Phase 3: Learn
Standardize ϕ_{cand} on $\mathcal{D}$
Select C via 5-fold CV on held-out log-loss
Fit ℓ_1-regularized logistic regression: $\mathbf{w}, b \leftarrow \arg\min \mathcal{L}_{\text{logistic}}(\mathbf{w}, b; \mathcal{D}) + \frac{1}{C}\|\mathbf{w}\|_1$
$\phi \leftarrow$ features with non-zero w_i
$g(s) \leftarrow \mathbf{1}[\sigma(\mathbf{w}^\top \phi(s) + b) > \tau]$
return g

C.1 Hyperparameters and Training Details

Phase 1 runs for $N_{\text{explore}} = 50$ episodes with $\varepsilon_{\text{explore}} = 0.5$. Before fitting in Phase 3, all features in ϕ_{cand} are standardized to zero mean and unit variance on $\mathcal{D}$, ensuring that the interpretable weight signs discussed in Section 4.3 are not confounded by scale differences across heterogeneous features. We induce sparsity using ℓ_1-regularized logistic regression with regularization strength C selected by 5-fold cross-validation over $C \in \{0.01, 0.03, 0.1, 0.3, 1.0, 3.0, 10.0\}$, optimizing held-out log-loss. Features with zero coefficient after fitting are dropped from the deployed

gate. The decision threshold τ in Eq. (5) is fixed to 0.5 for all environments.

C.2 Universal Feature Definitions

Table 6 defines the universal features available in every environment.

Table 6: Universal feature definitions.

Feature	Definition
`step_count`	Current step index t in the episode
`token_entropy`	Entropy of the next-token distribution
`action_entropy`	Entropy over candidate actions
`state_length`	Token count of the current state text

C.3 LLM Feature Layer

For environments where universal features may be inadequate, DIAL's optional LLM feature layer automates the proposal of task-specific signals. The LLM receives a structured summary of the exploration dataset $\mathcal{D}$, including: (1) aggregate statistics (total episodes, steps, optimizer trigger rate, overall positive-utility fraction), (2) per-step breakdowns of trigger rate and utility by step index, and (3) representative examples of states where the optimizer was triggered, paired with their utility labels (positive vs. negative).

Given this summary, the LLM is prompted to analyze patterns that distinguish positive-utility from negative-utility states and to produce a Python function that extracts exactly 5 task-specific features from the raw state text. Each feature must return a float value using only the Python standard library. The full prompt template is provided in §C.5.

The LLM-proposed features $\phi_{\text{LLM}}(s)$ are combined with the universal features to form the candidate pool $\phi_{\text{cand}}(s) = \phi_{\text{univ}}(s) \cup \phi_{\text{LLM}}(s)$. The ℓ_1-regularized gate (§4.3) then selects or discards each feature based on its predictive value for rollout utility. This two-stage design (LLM proposes, LASSO disposes) ensures that uninformative LLM features are automatically filtered out. As shown in Table 11, environments with richer task-specific structure (WebShop, TWExpress) retain more LLM features, while Plancraft retains none.

C.4 Online Adaptation

For drifting environments, DIAL can optionally refit during deployment using ε-greedy exploration ($\varepsilon : 0.1 \rightarrow 0$) and periodic retraining. During deployment, $\varepsilon_{\text{deploy}}$ decays from 0.1 to 0 over the first 100 episodes, and the gate is refit every $k = 30$ episodes using the accumulated ε-greedy samples. Refitting reuses the same solver configuration as the initial training, so no additional hyperparameter tuning is required at deployment time.

This mode is *not free*: ε-greedy overrides ignore the current gate, so in Type I environments a fraction of overridden steps will trigger harmful rollouts. For stationary environments, which are the setting of our main experiments, we recommend skipping online adaptation and relying on the offline-fitted gate.

C.5 LLM Reasoning Prompts

The full LLM prompt template is provided below. We use WebShop as the example, where LLM features contribute most meaningfully.

LLM Feature Layer Prompt Template

```
You are an AI agent that has
been exploring an interactive
environment.  During exploration,
you sometimes performed ``rollouts''
(additional computation) to improve
your decisions.  Sometimes the
rollout was useful (positive
utility), sometimes it was not.
Your task:  Analyze the patterns in
your experience and write a Python
function that extracts features from
the state text that could predict
whether a rollout would be useful.
```

[Exploration statistics and representative examples provided]

```
Requirements:
- Return a dict with string keys and
float values.
- Use only Python standard library.
- Extract exactly 5 features based
on the patterns you observe.
- Focus on features that DIFFER
between useful and not-useful
rollout cases.
```

Figure 6: Prompt template used in the LLM feature layer (Phase 2) to propose task-specific features from exploration data.

D Experimental Setup

D.1 Environment Specifications

We evaluate on 6 environments spanning diverse agent tasks. Table 7 summarizes their base performance, optimizer headroom, and optimizer type.

HotpotQA (Yang et al., 2018). A multi-hop question answering task where the agent must retrieve and reason over multiple Wikipedia paragraphs to answer complex questions. The agent iteratively issues search queries and reads returned passages before producing a final answer.

WebShop (Yao et al., 2022). A web navigation task where the agent shops for a product matching a natural language instruction. The agent navigates a simulated e-commerce website by searching, clicking product links, and selecting attributes (size, color) before purchasing.

FEVER (Thorne et al., 2018). A fact verification task where the agent must classify a claim as *Supported*, *Refuted*, or *Not Enough Info* based on evidence retrieved from Wikipedia. The agent searches for relevant passages and reasons over retrieved evidence to produce a verdict.

APPS (Hendrycks et al., 2021). A code generation task where the agent solves competitive programming problems. Given a problem description with input/output examples, the agent generates a Python solution and can iteratively debug based on test case feedback.

TWExpress (Jansen and Côté, 2023). A text-based game where the agent navigates a simulated environment to complete household tasks (e.g., finding and heating objects). The agent issues natural language commands (go, open, take, use) and observes textual descriptions of the resulting state.

Plancraft (Dagan et al., 2024). A crafting planning task where the agent must produce a target item by executing a sequence of crafting recipes in a Minecraft-inspired environment. The agent selects recipes and manages inventory to reach the goal item.

These environments cover the full range of the Two-Source Model: FEVER and HotpotQA are dominated by Type I states (information poverty), WebShop and APPS are dominated by Type D states (decision difficulty), TWExpress is rollout-safe ($\Delta{=}{+}31.8$), and Plancraft is rollout-harmful ($\Delta{=}{-}7.0$).

All three optimizer types rank candidates by environment-provided reward; they differ along two axes: what is varied (single action / full program / full action sequence) and how the rollout is aggregated (mean return of truncated rollouts / test-case pass / full-episode reward).

- **Per-action evaluation** (HotpotQA, FEVER, TWExpress, Plancraft). At the gated step, take the top-$K{=}5$ candidate actions proposed by the base policy. For each candidate a, fork the environment snapshot, execute a, and continue for $H{-}1{=}2$ further steps with the same LLM proposer ($\tau{=}0.7$), giving a truncated rollout of length $H{=}3$. Repeat $N{=}5$ independent rollouts per candidate and set $V(a) = \frac{1}{N}\sum_{n=1}^{N} R_n(a)$, where $R_n(a)$ is the env-provided score for the n-th truncated trajectory (task-specific: e.g., F1 for QA, episode return for games). Execute $a^* = \arg\max_a V(a)$.
- **K-variant sampling** (APPS). Sample $K{=}3$ complete code solutions and execute the first one that passes the provided test cases (objective check via the interpreter).
- **LLM-Propose-K** (WebShop). The LLM proposes $K{=}3$ full action sequences; each is simulated end-to-end in the environment, and the trajectory with the highest observed episodic reward is executed.

Utility estimation. In all environments, the utility label U_t used for gate training (§4.1) is computed by paired counterfactual evaluation from the same state snapshot. For per-action evaluation environments, the base policy's proposed action a_B is included in the top-K candidate set and evaluated with the same $N{\times}H$ rollout protocol as all other candidates; utility is $U_t = \mathbf{1}[V(a^*) > V(a_B)]$, where $V(a) = \frac{1}{N}\sum_{n=1}^{N} R_n(a)$. For K-variant sampling (APPS), the base solution is evaluated on the same test suite as the K variants, and $U_t = \mathbf{1}[\max_k \text{pass_rate}_k > \text{pass_rate}_{\text{base}}]$. Because all candidates are evaluated from identical initial conditions, this is a counterfactual estimate rather than an on-policy observation. The truncation horizon H introduces approximation bias relative to full-episode returns; we find results stable for $H \in [3, 10]$.

In all three cases, the optimizer itself has access to the env's scoring function; our method is about learning *when* this optimizer is worth invoking, not how it scores internally.

Table 7: Environment setup. Base/Always: SR without/with optimizer. Δ: rollout headroom.

Environment	Base SR	Always SR	Δ	Optimizer T
HotpotQA	49.0%	97.0%	+48.0	Per-action eval
WebShop	7.2%	43.0%	+35.8	LLM-Propose-K
FEVER	37.0%	99.8%	+62.8	Per-action eval
TWExpress	67.5%	99.3%	+31.8	Per-action eval
Plancraft	29.8%	22.8%	−7.0	Per-action eval
APPS	60.5%	79.5%	+19.0	K-variant sampling

D.2 Detailed Baseline Comparison

Table 8 provides a detailed comparison of all baselines evaluated in this paper. All six methods share a common assumption: a fixed mapping from signal value to compute need.

D.3 Computational Cost Breakdown

Table 9 breaks down the per-method cost structure. Cost is reported as ×base: total *deployment* tokens normalized by base_only tokens per episode. Calibration or exploration overhead is excluded for all methods, ensuring a fair comparison of runtime efficiency. This metric captures hidden per-step overhead that raw rollout counts miss: CATTS incurs $K{=}5$ forward passes every step regardless of the gating decision, and AUQ issues a confidence query every step. Consistent with the cost accounting of §4, DIAL incurs a one-time exploration cost of $N_{\text{explore}}{=}50$ episodes per (environment, backbone) deployment and sub-second offline gate fitting; at deployment the gate adds zero per-step inference cost beyond the feature evaluation and a single sigmoid: no extra LLM call, no K-way voting, and no calibration query.

Table 9: Computational cost breakdown per method. Gate OH = additional LLM calls per step beyond the base proposer. Cost (×base) = total *deployment* tokens normalized by base_only tokens per episode, averaged across 6 environments. Calibration/exploration overhead is excluded for all methods. †Requires separate calibration data.

Method	Gate OH/step	OH source	Avg ×b
base_only	0	—	1.00
always_trigger	0	—	6.00
CATTS	$+K{=}5$ calls	Voting (fwd pass)	7.98
SEAG†	0	Reads logprob	6.37
CoRefine†	0	Reads entropy	6.33
CaTS†	0	Reads confidence	8.26
AUQ	+1 query	Confidence question	7.67
s1_budget	0	—	6.35
DIAL	**0**	**Learned gate**	**5.85**

E Full Results

Table 10 reports SR and Cost (×base) for all methods across 6 environments and 3 backbones. Cost includes all deployment LLM calls (base proposer, gate overhead, rollouts) normalized by base_only cost. Bounds comparison is in Table 2 (§5.2). Several patterns are consistent across backbones.

Qwen3-4B. DIAL achieves the highest SR in 4 of 6 environments (HotpotQA, WebShop, TWExpress, APPS). The ×base metric reveals hidden costs: CATTS on APPS costs 6.00×base (due to $K{=}5$ per-step voting) despite barely triggering rollouts, while DIAL costs only 2.61× at 12.2 pp higher SR. FEVER is the main exception: DIAL achieves the highest SR (49.8%) but at 16.51×base, the highest cost among all environments. This failure mode is analyzed in §6.

Phi-3.5-mini. Fixed-direction baselines degrade sharply on this backbone. On FEVER, all six baselines score at or below 20% (base: 7.2%), while DIAL reaches 20.5%. On WebShop, DIAL (57.3%) exceeds the best baseline (s1_budget, 43.0%) by 14.3 pp at lower cost (2.56× vs. 3.89×).

Llama-3.1-8B. Direction reversal interacts with model capacity. On TWExpress, CaTS collapses to near-base performance (35.8% vs. base 36.5%) because it rarely triggers, while DIAL achieves 94.8%. On FEVER, DIAL (54.7%) is competitive with CaTS (56.3%), unlike Qwen3 where FEVER is the weakest environment. On APPS, AUQ drops to 46.0% (below base 60.2%), demonstrating that wrong-direction gating actively harms performance on this backbone.

F Signal and Feature Analyses

This appendix collects empirical analyses of signal identity and signal sufficiency referenced in Section 3.1.

F.1 Per-Environment Feature Selection

The LASSO coefficient matrix identifies, per environment, which auxiliary features carry information about the latent state type τ, the components of $\phi(s)$ that lift σ toward sufficiency for VOC, as predicted by App B.1. `step_count` is informative in 4 of 5 environments with significant signal: it proxies *information accumulation*, distinguishing late-step Type I states (where information has accumulated to a known shortage) from early-step Type D states (where multiple options are still viable). WebShop is the exception, relying on `num_available_actions` and task-specific features (`price_mentioned`, `action_is_click`); these proxy *decision optionality* directly, as expected for a choice-dominated (Type D) environment. In every case, the features LASSO retains are interpretable as proxies for the two-source decomposition rather than ad-hoc engineering choices.

Table 8: Detailed comparison of baselines. All methods assume a fixed signal–utility direction. †Requires a separate labeled dataset for calibration.

Method	Signal	Direction	Granularity	Mechanism
CATTS (Lee et al., 2026)	Vote entropy + margin	High disagreement → trigger	Per-step	K=5 forward passes, arbiter on disagreement
SEAG† (Lee et al., 2025)	Mean token confidence	Low conf. → search	Per-problem	Platt scaling, search depth allocation
CaTS† (Huang et al., 2026a)	Calibrated confidence	Low conf. → generate	Per-attempt	Self-calibration, Best-of-N early stopping
CoRefine† (Jin et al., 2026)	Conv1D confidence	Low conf. → refine	Per-problem	Halt / re-examine / redirect controller
s1_budget (Muennighoff et al., 2025)	Token budget	Fixed budget → stop	Per-problem	Budget-constrained generation
AUQ (Zhang et al., 2026)	Verbalized uncertainty	High uncert. → reflect	Per-step	Uncertainty-aware memory + reflection
DIAL	**Multi (auto)**	**Learned**	**Per-step**	Explore + reason + direction learning

Table 10: Full results across 6 environments and 3 backbones. SR (%) and Cost (×base, total deployment tokens normalized by base_only). Best and second-best SR per environment. †: methods requiring calibration data. Cost includes all LLM calls: base proposer, gate overhead (CATTS K=5 voting, AUQ confidence query), and rollout calls.

	HotpotQA		**WebShop**		**FEVER**		**TWExpress**		**Plancraft**		**APPS**	
Method	SR↑	Cost↓	SR↑	Cost↓	SR↑	Cost↓	SR↑	Cost↓	SR↑	Cost↓	SR↑	Cost↓
Qwen3-4B-Instruct												
CaTS†	93.2	10.54	30.5	3.45	48.2	25.48	96.7	3.37	20.3	4.14	66.2	2.61
SEAG†	67.5	9.34	28.0	2.84	47.3	17.19	97.3	2.60	22.8	3.52	66.0	2.75
CoRefine†	68.2	9.28	27.5	2.78	47.8	17.19	97.5	2.57	20.8	3.44	67.5	2.70
CATTS	68.3	10.53	16.0	5.55	32.2	16.46	97.5	2.83	23.0	6.53	60.8	6.00
AUQ	95.0	10.29	35.7	5.83	38.7	18.04	95.5	1.95	22.2	6.64	64.7	3.28
s1_budget	94.0	8.27	17.8	2.91	44.2	19.21	95.0	2.02	16.3	3.02	69.0	2.65
DIAL	**95.2**	8.02	**43.8**	2.50	**49.8**	16.51	**99.0**	1.81	**23.3**	3.63	**73.0**	2.61
Phi-3.5-mini-Instruct												
CaTS†	68.3	5.86	41.7	2.52	19.8	14.51	68.2	1.00	13.5	1.44	30.5	3.55
SEAG†	88.3	7.82	36.5	2.43	13.5	5.61	92.5	2.58	14.7	2.29	27.8	2.59
CoRefine†	87.7	7.83	35.8	2.44	13.7	5.73	92.7	2.56	15.2	2.24	28.5	2.60
CATTS	42.0	7.98	28.7	4.86	12.7	7.65	86.7	4.00	13.5	5.96	27.7	6.10
AUQ	31.2	5.47	37.0	3.11	8.5	3.18	92.5	1.82	12.7	2.24	27.2	2.03
s1_budget	91.2	5.26	43.0	3.89	20.0	8.45	93.7	2.05	13.7	1.73	34.5	2.93
DIAL	**92.3**	5.38	**57.3**	2.56	**20.5**	7.44	**96.7**	1.83	**16.8**	1.33	**36.8**	2.87
Llama-3.1-8B-Instruct												
CaTS†	94.2	7.21	36.0	2.51	**56.3**	14.82	35.8	3.00	22.8	3.42	55.2	4.81
SEAG†	94.0	9.82	27.5	1.86	53.5	12.57	76.5	3.85	21.8	2.70	57.2	4.41
CoRefine†	94.2	9.42	27.5	1.82	53.5	12.64	76.0	3.56	21.5	2.69	57.2	4.37
CATTS	94.0	11.11	11.3	4.74	14.2	17.00	49.3	4.11	27.0	7.31	52.3	6.70
AUQ	95.0	10.54	37.8	4.54	52.8	25.08	63.2	3.58	27.0	8.21	46.0	6.11
s1_budget	94.5	8.08	11.2	1.87	13.8	8.56	76.1	3.05	26.3	2.53	52.5	2.33
DIAL	**95.5**	7.07	**41.7**	1.68	54.7	18.69	**94.8**	2.55	**27.0**	2.49	**59.7**	4.29

F.2 LLM-Generated Features

Table 11 reports the features generated by the LLM reasoning step (Phase 2) for each environment. The LLM receives exploration statistics and representative positive/negative examples, then produces a Python function extracting 5 task-specific features (the full prompt template is in §C.5). LASSO subsequently selects or discards each feature based on its predictive value for rollout utility.

Environments with rich task-specific structure benefit most from LLM features: WebShop's choice-dominated interface is well captured by `price_mentioned` and `action_is_click`, while TWExpress's exploration mechanics are reflected in `closed_ratio` and `action_look_around`. In contrast, Plancraft's rollout-harmful nature means no LLM feature improves prediction, and LASSO correctly filters all five proposals. This pattern aligns with the LLM ablation results (Table 4): the environment where LLM features contribute most to SR (FEVER, +9.1 pp) is precisely where the LLM discovers unique signals (`has_claim`) that the universal feature pool lacks.

Figure 7 shows the complete LLM-generated feature extraction function for WebShop (seed 42), the environment with the most LLM features retained.

LLM Output for WebShop

```
def extract_features(state_text, step_count,
     action_text):
   import re
   features = {}
   # Instruction keyword density
   keywords = ["gluten free", "protein",
      "price lower than", "size", "color"]
   matches = sum(1 for kw in keywords
      if kw in state_text.lower())
   features['instruction_keyword_count'] = (
      float(matches) / len(keywords))
   # Early-step indicator
   features['step_early'] = (
      1.0 if step_count <= 2 else 0.0)
   # Click vs. search action
   features['action_is_click'] = (
      1.0 if action_text.startswith("click[")
      else 0.0)
   # Price constraint presence
   price_matches = len(re.findall(
      r"price lower than|price:|\$",
      state_text, re.IGNORECASE))
   features['price_mentioned'] = (
      float(price_matches) / 5.0)
   # Filter complexity
   filter_kws = ["size","color","flavor",
      "price","rating"]
   features['filter_count'] = sum(
      1 for kw in filter_kws
      if kw in state_text.lower()) / 5.0
   return features
```

Figure 7: Complete LLM-generated feature extraction function for WebShop (seed 42), the environment with the most LLM features retained by LASSO.

F.3 AUC Hierarchy: Empirical Signature of σ-Insufficiency

Table 12 is the empirical instantiation of the sufficient-statistic claim of §3.1: it measures, on held-out data, how much of VOC each gate family can recover. Single entropy is near chance (AUC≈0.50) in every environment, the operational signature of σ's insufficiency for VOC. Multi-signal logistic gates that augment σ with universal auxiliary features reach 0.74–0.92, recovering most of the available signal; hidden-state probes (which have access to the model's full internal representation, an upper bound on what any feature map $\phi(s)$ can extract) reach 0.79–0.99. The gap between the single-entropy row and the multi-signal row is therefore a quantitative measure of how much auxiliary conditioning is required to lift σ toward sufficiency for VOC; the residual gap between multi-signal and hidden-state probe is the sufficiency that universal features fail to capture (small in most environments). The bottleneck is signal *combination*, not gate *capacity*, and the multi-signal LR row is the empirical case for DIAL's design.

Table 12: AUC by gate family. Single entropy is near chance; multi-signal gates close most of the gap to hidden-state probes. FEVER and TWExpress are omitted because their short episodes yield insufficient held-out data for reliable AUC estimation.

Gate family	HotpotQA	APPS	WebShop	Plancraft
Single entropy	0.50	0.56	0.50	0.50
Best single signal	0.78	0.78	0.90	0.74
Multi-signal LR	0.85	0.76	0.92	0.74
Hidden-state probe	0.87	0.79	0.99	0.95

G Robustness and Additional Experiments

This appendix collects Two-Source Model verification data, robustness checks for direction reversal, and additional experimental analyses.

G.1 Temporal Dynamics (P1)

Table 13 reports the full early/late ρ data referenced in Figure 5. The Two-Source Model predicts that $\rho(\text{entropy}, U)$ should decrease from

Table 11: LLM-generated features per environment. Each environment receives 5 LLM-proposed features; LASSO selects those with non-zero coefficients. Environments with richer task-specific structure (WebShop, TWExpress) retain more LLM features.

Environment	Sel./Gen.	LASSO-Selected LLM Features
WebShop	4/5	`price_mentioned`, `action_is_click`, `step_early`, `instruction_keyword_count`
TWExpress	4/5	`text_length`, `closed_ratio`, `action_look_around`, `already_open`
HotpotQA	3/5	`has_question`, `entity_count`, `fact_keyword_count`
FEVER	2/5	`text_length_normalized`, `has_claim`
APPS	1/5	`action_type`
Plancraft	0/5	(none selected)

early to late steps within each episode, as late-step uncertainty increasingly reflects the residual Type I component after information-gathering attempts.

Table 13: Temporal shift in ρ(entropy, U). All environments show ρ shifting more negative from early to late steps, consistent with P1. Plancraft omitted ($|\rho|$=0.016, too weak for meaningful comparison).

Env	Phase	ρ	95% CI	n	p	Shift
HotpotQA	early	-0.176	$[-0.260, -0.091]$	509	<0.001	$\Delta=-0.261$
	late	-0.437	$[-0.500, -0.370]$	699	<0.001	
APPS	early	$+0.102$	$[-0.013, +0.209]$	290	0.084	$\Delta=-0.246$
	late	-0.144	$[-0.281, +0.001]$	199	0.043	
WebShop	early	$+0.285$	$[+0.222, +0.346]$	600	<0.001	$\Delta=-0.291$
	late	-0.006	n.s.	473	0.895	
FEVER	early	$+0.054$	n.s.	200	0.446	$\Delta=+0.024$
	late	$+0.078$	n.s.	82	0.486	
TWExpress	early	$+0.161$	—	422	<0.001	$\Delta=-0.153$
	late	$+0.008$	n.s.	376	0.877	

The prediction holds across 4 of 5 testable environments: HotpotQA shows the strongest shift ($\Delta=-0.261$), with late-step entropy reflecting deeper information poverty after failed evidence retrieval. WebShop's strong positive early-phase signal ($\rho=+0.285$) vanishes in the late phase ($\rho=-0.006$, n.s.), consistent with the choice-dominated structure being resolved early. In APPS, ρ reverses from marginally positive ($+0.102$) to negative (-0.144), as late-step decision complexity gives way to residual information poverty.

FEVER is the exception: both phases show weak, non-significant positive ρ, consistent with FEVER's short episodes (median 8 steps) concentrating most rollout value at step 0–1, leaving insufficient late-step variation for a meaningful shift.

G.2 Signal Identity Alignment (P3)

Table 14 reports the strongest signal per environment and its interpretation under the Two-Source Model. In Type I environments, information-sufficiency proxies (`step_count`) dominate; in mixed/Type D environments, decision-complexity proxies (`num_available_actions`) dominate.

Table 14: P3 Signal identity alignment. The strongest signal in each environment matches the Two-Source Model prediction: information-sufficiency proxies dominate in Type I environments; decision-complexity proxies dominate in mixed/Type D.

Env	Type	Strongest Signal	$\lvert\rho\rvert$	Interpretation
HotpotQA	Info-Poverty	step_count	0.494	Info accumulation
FEVER	Info-Poverty	step_count	0.619	Info accumulation
TWExpress	Info-Poverty	step_count	0.477	Exploration depth
WebShop	Mixed	num_avail_actions	0.444	Decision-space size
APPS	Decision-Diff	step_count	0.339	Debugging depth
Plancraft	Weak (harmful)	has_output	0.162	Rollout harmful

G.3 Multi-Backbone Verification

We validate our core findings on three backbones from three vendors: Qwen3-4B-Instruct (Alibaba, 4B), Phi-3.5-mini-instruct (Microsoft, 3.8B), and Llama-3.1-8B-Instruct (Meta, 8B) across all 6 environments. Table 10 reports full results. This appendix supports the (environment, backbone)-indexed formulation of §3.2: the same task changes p_I (and, when it crosses p_I^*, the sign of ρ) when the backbone changes, so backbone-level reversal is a prediction of the two-source model rather than a confound to be controlled for.

Figure 8 visualizes the direction variation, with gold borders marking sign flips. On Phi-3.5 FEVER: most fixed-direction baselines score near or below base_only (7.2% base $\rightarrow$ 8.5–23%), demonstrating that the fixed-direction assumption is brittle when the model changes (Figure 9).

G.4 Stratified Analysis: Controlling for Trajectory Length

Trajectory length varies across environments and correlates with both entropy and utility. To rule out length as a confounder, we stratify by step count and recompute ρ(entropy, U) within each stratum. Direction reversal persists within trajectory-length

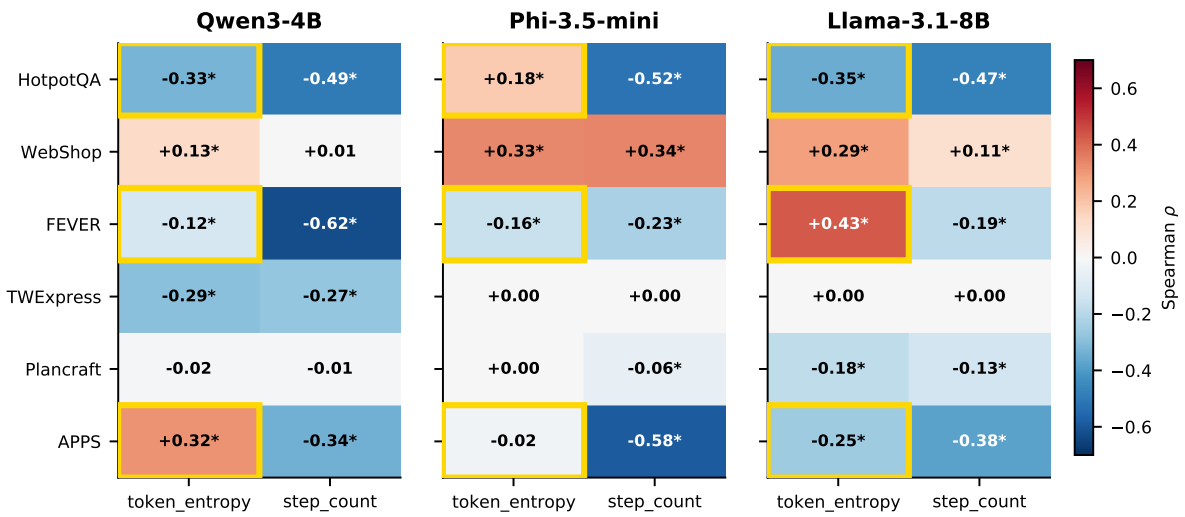

Figure 8: Multi-backbone signal direction heatmap. Gold borders mark cells where the sign of ρ flips across backbones with significant correlations. Entropy direction is model-dependent; structural signals (step_count) are more stable.

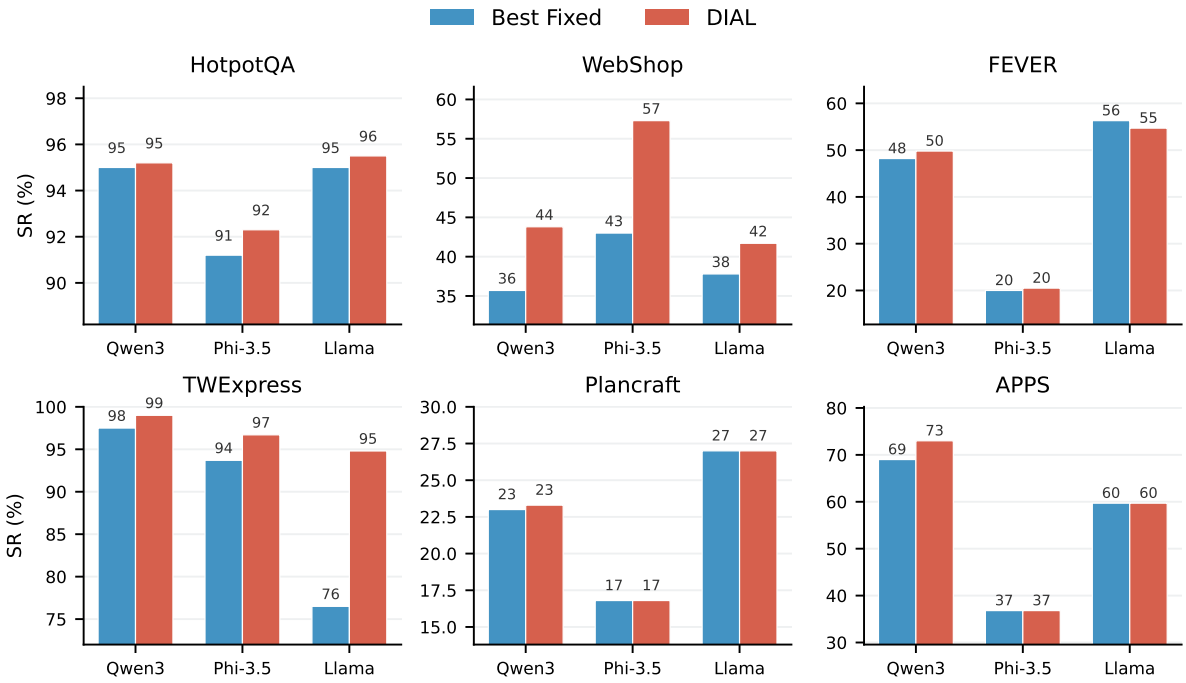

Figure 9: Cross-backbone comparison: DIAL (red) vs. best fixed-direction baseline (blue) across 3 backbones. Fixed baselines show large SR variance across backbones; DIAL adapts robustly.

strata where utility variance is non-zero. In HotpotQA, ρ remains negative across all three strata ($-0.18/-0.46/-0.42$). The reversal is not an artifact of trajectory length.

G.5 Regularizer Ablation

The signed-weight diagnostic of §4.3 and DIAL's robustness to noisy LLM-proposed features (§4.2) both rely on the *exact* sparsity induced by the ℓ_1 penalty in Eq. (4). To verify that ℓ_1 specifically, rather than any shrinkage or selection mechanism, is responsible for these properties, we ablate the regularizer while holding all other pipeline components fixed: same exploration dataset $\mathcal{D}$, same candidate feature pool ϕ_{cand}, same 5-fold CV protocol for hyperparameters, same threshold τ=0.5. We compare five variants:

- ℓ_1**-logistic** (DIAL default): joint sparsity and shrinkage.
- ℓ_2**-logistic** (Ridge): shrinkage only; all features retained with small non-zero weights.
- **No regularization**: plain logistic fit on ϕ_{cand}.
- **Elastic Net** (α=0.5, mixing ℓ_1 and ℓ_2).
- **MI top-k + unregularized**: hard feature selection by mutual information (top-3), then unregularized logistic on the retained features. This variant isolates *selection* from *shrinkage*.

Table 15 reports SR and the mean number of non-zero fitted weights across three environments spanning signal strength: HotpotQA ($|\rho|$=0.494, strong), WebShop ($|\rho|$=0.444, moderate), and Plancraft ($|\rho|$=0.016, weak).

Table 15: Regularizer ablation on three environments ordered by signal strength. SR (%); "#nz" is the mean number of non-zero fitted weights across three seeds. The candidate pool ϕ_{cand} contains 31 features (5 universal $+3$ derived $+20$ hidden-state PCA $+3$ auto-extracted); adding the LLM feature layer brings the pool to $\approx$36. ℓ_1 is the only variant that achieves competitive SR *and* exact sparsity required by the signed-weight diagnostic.

	HotpotQA $\lvert\rho\rvert$=0.494		WebShop $\lvert\rho\rvert$=0.444		Plancraft $\lvert\rho\rvert$=0.016	
Regularizer	SR	#nz	SR	#nz	SR	#nz
ℓ_1 (DIAL)	95.2	7.3	43.8	8.3	23.3	1.7
ℓ_2	TBD	36	TBD	36	TBD	36
No regularization	TBD	36	TBD	36	TBD	36
Elastic Net	TBD	TBD	TBD	TBD	TBD	TBD
MI top-3 + unreg.	TBD	3	TBD	3	TBD	3

Note. ℓ_1 #nz values are measured across three seeds of DIAL's production runs; the per-environment mean reveals adaptive sparsity: under strong signal (HotpotQA, WebShop) DIAL retains $\approx$7–8 features, whereas under weak signal (Plancraft) ℓ_1 collapses the gate to $\approx$2 features, absorbing the remainder into the bias and yielding a near-constant trigger probability consistent with the low rollout headroom of that environment (Figure 10).

The comparison isolates three hypotheses underlying DIAL's regularizer choice.

ℓ_1 vs. ℓ_2: sparsity is required for diagnosis, not just for predictive performance. Both penalties shrink noisy coefficients toward zero, so on strong-signal environments their gating SRs are expected to be comparable. The critical distinction lies in interpretability: under ℓ_2, every feature retains a small non-zero weight, collapsing the clean {informative, uninformative} partition of §4.3 into a continuous ranking. The signed-weight summary no longer reads off which signals the environment treats as informative; it merely ranks them. Reports of "non-zero weights per environment" in the main text (e.g., the direction analyses of Table 3) are only interpretable under exact sparsity.

ℓ_1 vs. unregularized logistic: robustness under weak signals. On weak-signal environments like Plancraft, the unregularized fit is expected to latch onto sample-level noise: with $|\rho|\approx$0.02, any 250-sample fit without penalty will assign large weights to whichever feature happens to pseudo-correlate on the training split. ℓ_1's shrinkage absorbs this noise into the bias term, producing a gate that triggers conservatively, consistent with the observed low trigger rate in Plancraft (Figure 10).

ℓ_1 vs. MI top-k + unregularized: joint selection+shrinkage vs. selection alone. Hard feature selection by mutual information separates the two roles ℓ_1 plays. The resulting gate is expected to recover most of ℓ_1's SR on strong-signal environments, but exhibits two practical drawbacks: (i) k must be tuned per environment rather than emerging from CV over a single hyperparameter C, reintroducing the per-environment engineering cost that DIAL aims to eliminate; and (ii) the retained features lose the shrinkage calibration that stabilizes weights near the decision boundary, making the gate more sensitive to τ.

Summary. The three comparisons establish that ℓ_1 is not interchangeable with other shrinkage or selection schemes. It is the minimal choice that simultaneously delivers competitive SR, the *exact* sparsity required by the signed-weight diagnostic, and robustness to noisy feature proposals from the optional LLM layer.

G.6 Gate Complexity Ablation

To verify that the bottleneck is direction rather than gate complexity, we compare gate architectures with correct vs. wrong direction (Table 16).

Table 16: Gate complexity ablation on HotpotQA. Direction matters more than model capacity: a logistic gate with correct direction outperforms an MLP with wrong direction by 49.9 pp.

Gate	Direction	SR (%)	Δ
Logistic (5 feat)	Correct	95.2	+46.2
MLP (5 feat)	Correct	∼95	∼+46
Hidden state (2560-d)	Correct	∼95	∼+46
Logistic (5 feat)	Wrong	62.0	+13.0
MLP (5 feat)	Wrong	45.3	−3.7
No gate (base)	n/a	49.0	0

With correct direction, all architectures achieve ∼95% SR; the jump from logistic to MLP to hidden-state probe adds <1 pp. With wrong direction, higher capacity makes performance *worse* (MLP 45.3% < logistic 62.0%), because the more powerful gate more precisely targets harmful states. This confirms that signal direction consistently outweighs gate complexity, and explains why DIAL's minimal gate design suffices.

G.7 Environment-Adaptive Trigger Behavior

Beyond direction and feature selection, DIAL's gating magnitude also adapts to each environment without explicit programming. Figure 10 shows the per-step trigger rate across all 6 environments, ordered by rollout headroom Δ. In rollout-safe TWExpress (Δ=+31.8), DIAL triggers aggressively (RR = 73%), consistent with the high headroom. In rollout-harmful Plancraft (Δ=−7.0), the trigger rate decays from 49% at step 0 to <20% in later steps, as the gate learns that late-step rollouts are increasingly harmful. WebShop and APPS show moderate, stable trigger rates. This full behavioral range emerges from the per-environment signed weights fitted in §4.3: the gate does not need an explicit headroom estimate, because signal direction already encodes whether the environment rewards or penalizes additional computation.

H Broader Impact

This work addresses computational efficiency in LLM agent deployment. By reducing unnecessary optimizer invocations, DIAL lowers the energy and cost footprint of test-time compute without sacrificing task performance. The prescriptive framework (characterize information structure before designing gates) may help practitioners avoid deploying fixed-direction methods that actively harm performance in mismatched environments. We do not foresee direct negative societal impacts beyond those inherent to the LLM agents themselves.

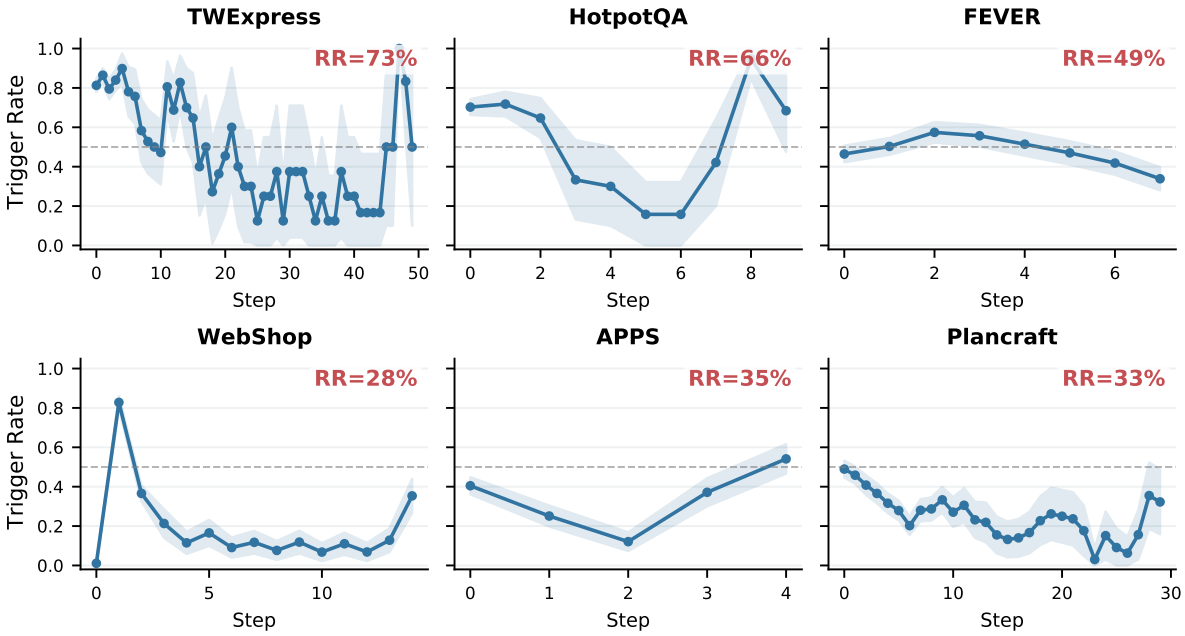

Figure 10: Environment-adaptive trigger behavior. Panels ordered by rollout headroom Δ. Solid line: per-step trigger rate with 95% CI band. Dashed gray: 50% reference. RR: overall trigger rate. DIAL's trigger rate ranges from 73% (TWExpress) to <20% (late-step Plancraft), emerging from direction learning alone.